DSCC/CSC/TCS 462 Project

Group 3

2025-12-02

Contents

#Data Pre-processing:

In this step, we want to:

1. Read the data

```
ev_data <- read.csv("Data/EV_Charging/EV_Charging_Data-_Municipal_Lots_and_Garages_20251
    stringsAsFactors = FALSE, check.names = FALSE)
```

2. Choose ID tag as an unique identifier:

```
# Choose ID_tag as unique identifier Check if ID_tag is
# unique
sum(duplicated(ev_data$`ID Tag`))
```

```
## [1] 176776
```

```
# Keep only the first row per ID_tag
ev_data <- ev_data %>%
    distinct(`ID Tag`, .keep_all = TRUE)
```

3. Remove Columns and rows that are not relevent to our experiment:

```
# Remove columns: Driver_id, Invalidity_reason, country
ev_data <- ev_data %>%
    select(-`Driver ID`, -`Invalidity Reason`, -Country)
```

```r
# Remove rows where session_status is "Disconnected"
ev_data <- ev_data %>%
    filter(`Session Status` != "DISCONNECTED")

# Keep Station_name and drop Charger_id
ev_data <- ev_data %>%
    select(-`Charge Box ID`)

# Check null values and drop rows with any NA
colSums(is.na(ev_data))
```

```
##                      Date              Station Name             Location Name
##                         0                         0                         0
##              Connector ID                     ID Tag           Connected Time
##                         0                         0                         0
##         Disconnected Time    Charge Duration (min) Connected Duration (min)
##                         0                         0                         0
##    Energy Provided (kWh)            Session Status
##                         0                         0
```

```r
ev_data <- ev_data %>%
    tidyr::drop_na()
```

4.Convert type from chr to numeric

```r
# change Charge Duration (min) to numeric
ev_data <- ev_data %>%
    mutate(`Charge Duration (min)` = as.numeric(gsub(",", "",
        `Charge Duration (min)`)))
# change Connected Duration (min) to numeric
ev_data <- ev_data %>%
    mutate(`Connected Duration (min)` = as.numeric(gsub(",",
        "", `Connected Duration (min)`)))
```

5. Add columns based on Dates

```r
# Sort the data based on date
ev_data$Date <- as.Date(ev_data$Date, format = "%m/%d/%Y")
ev_data <- ev_data %>%
  arrange(Date)

# Add weekday column
```

```r
ev_data <- ev_data %>%
  mutate(weekday = weekdays(Date))

# Add is_weekend
weekends <- c("Saturday", "Sunday")
ev_data <- ev_data %>%
  mutate(
    is_weekend = weekday %in% weekends
  )

# Add holidays column for year 2024
# Need to work on this!!
holidays_2024 <- as.Date(c(
  "2024-01-01", # New Year's Day
  "2024-01-15", # MLK Day
  "2024-02-19", # Presidents' Day
  "2024-05-27", # Memorial Day
  "2024-06-19", # Juneteenth
  "2024-07-04", # Independence Day
  "2024-09-02", # Labor Day
  "2024-10-14", # Columbus/Indigenous Peoples' Day
  "2024-11-11", # Veterans Day
  "2024-11-28", # Thanksgiving
  "2024-12-25"  # Christmas
))

ev_data <- ev_data %>%
  mutate(
    is_holiday_2024 = Date %in% holidays_2024
  )
```

6. Clean the location names:

```r
# Standardize location names
ev_data <- ev_data %>%
  mutate(
    `Location Name` = case_when(
      # Jerome Gun Hill Road variations
      grepl("JGU|Jerome.*Gun Hill", `Location Name`, ignore.case = TRUE) ~
        "Jerome Gun Hill Road Municipal Parking Garage",

      # Bay Ridge variations
      grepl("BRI|Bay Ridge", `Location Name`, ignore.case = TRUE) ~
        "Bay Ridge Municipal Parking Garage",
```

```
      # Delancey and Essex variations
      grepl("DES|Delancey.*Essex", `Location Name`, ignore.case = TRUE) ~
        "Delancey and Essex Municipal Parking Garage",

      # Queens Borough Hall variations
      grepl("QBO|Queensboro Hall|Queens Borough Hall", `Location Name`, ignore.case = TR
        "Queens Borough Hall Municipal Parking Garage",

      # Jerome 190th Street variations
      grepl("JON|Jerome.*190", `Location Name`, ignore.case = TRUE) ~
        "Jerome 190th Street Municipal Parking Garage",

      # St. George Courthouse variations
      grepl("SGE|St\\.? George", `Location Name`, ignore.case = TRUE) ~
        "St. George Courthouse Garage",

      # Court Square variations
      grepl("CSQ|Court Square", `Location Name`, ignore.case = TRUE) ~
        "Court Square Municipal Parking Garage",

      # Queens Family Court variations
      grepl("QFA|Queens Family Court", `Location Name`, ignore.case = TRUE) ~
        "Queens Family Court Municipal Garage",

      # Handle N/A
      `Location Name` == "N/A" ~ NA_character_,

      # Keep everything else as is
      TRUE ~ `Location Name`
    )
  )

# Check the results
unique(ev_data$`Location Name`)
```

```
## [1] "Delancey and Essex Municipal Parking Garage"
## [2] "Court Square Municipal Parking Garage"
## [3] "Jerome 190th Street Municipal Parking Garage"
## [4] "Bay Ridge Municipal Parking Garage"
## [5] "St. George Courthouse Garage"
## [6] "Queens Family Court Municipal Garage"
## [7] "Queens Borough Hall Municipal Parking Garage"
## [8] "Jerome Gun Hill Road Municipal Parking Garage"
## [9] NA
```

```r
# Count how many of each location
ev_data %>% count(`Location Name`, sort = TRUE)
```

```
##                                  Location Name    n
## 1   Delancey and Essex Municipal Parking Garage 9030
## 2         Court Square Municipal Parking Garage 8357
## 3  Queens Borough Hall Municipal Parking Garage 3228
## 4  Jerome 190th Street Municipal Parking Garage  832
## 5            Bay Ridge Municipal Parking Garage  732
## 6                 St. George Courthouse Garage  640
## 7          Queens Family Court Municipal Garage  344
## 8 Jerome Gun Hill Road Municipal Parking Garage  220
## 9                                          <NA>    8
```

7. Data info:

```r
head(ev_data, 10)
```

```
##          Date Station Name                                Location Name
## 1  2021-07-31      EVB0301  Delancey and Essex Municipal Parking Garage
## 2  2021-07-31      EVB0302  Delancey and Essex Municipal Parking Garage
## 3  2021-07-31      EVB0305        Court Square Municipal Parking Garage
## 4  2021-07-31      EVB0415 Jerome 190th Street Municipal Parking Garage
## 5  2021-07-31      EVB0415 Jerome 190th Street Municipal Parking Garage
## 6  2021-07-31        NYC19           Bay Ridge Municipal Parking Garage
## 7  2021-07-31   NYCDOT DE2  Delancey and Essex Municipal Parking Garage
## 8  2021-07-31   NYCDOT DE3  Delancey and Essex Municipal Parking Garage
## 9  2021-08-01      EVB0294           Bay Ridge Municipal Parking Garage
## 10 2021-08-01      EVB0297        Court Square Municipal Parking Garage
##    Connector ID              ID Tag Connected Time Disconnected Time
## 1             1 25JES6HTLHYJZZJ7IZBP       19:12:43          21:45:13
## 2             1 AE6032D1R8SRN0OUQ25D       22:32:32          07:44:35
## 3             1 WZTIYJAVUDXQ9S8YPT6F       23:03:26          11:08:02
## 4             1           gHIPSZS6I5       09:58:31          11:25:40
## 5             1 GXS7PP6E3I9I7VMCG9N8       12:02:16          15:08:23
## 6             1 UKYOL0P8H8V5RWTNTR1N       13:28:25          05:17:19
## 7             1 96P3P12RR3ECT6LHEYR0       19:42:05          17:00:42
## 8             1           5fwg1oJTG6       10:25:35          20:40:03
## 9             1             A6CE494E       12:50:52          06:24:20
## 10            1 Q0TNUAGIQVBQER6VVC61       10:49:21          17:36:38
##    Charge Duration (min) Connected Duration (min) Energy Provided (kWh)
## 1              152.5000                152.5000                16.251
## 2              552.0500                552.0500                57.683
```

```
## 3               290.1667                724.6000                31.030
## 4                87.1500                 87.1500                 9.301
## 5               135.4167                186.1167                10.357
## 6               325.0462                948.9072                31.333
## 7              1275.2539               1278.6044                50.745
## 8               230.0212                614.4603                22.547
## 9               140.6333               1053.4667                13.318
## 10              407.2833                407.2833                45.347
##    Session Status  weekday is_weekend is_holiday_2024
## 1            PAID Saturday       TRUE           FALSE
## 2            PAID Saturday       TRUE           FALSE
## 3            PAID Saturday       TRUE           FALSE
## 4            PAID Saturday       TRUE           FALSE
## 5            PAID Saturday       TRUE           FALSE
## 6            PAID Saturday       TRUE           FALSE
## 7            PAID Saturday       TRUE           FALSE
## 8            PAID Saturday       TRUE           FALSE
## 9            PAID   Sunday       TRUE           FALSE
## 10           PAID   Sunday       TRUE           FALSE
```

```
# Print column names nicely
cat("Column names:\n")
```

```
## Column names:
```

```
cat(paste(1:length(names(ev_data)), names(ev_data), sep = ". "),
    sep = "\n")
```

```
## 1. Date
## 2. Station Name
## 3. Location Name
## 4. Connector ID
## 5. ID Tag
## 6. Connected Time
## 7. Disconnected Time
## 8. Charge Duration (min)
## 9. Connected Duration (min)
## 10. Energy Provided (kWh)
## 11. Session Status
## 12. weekday
## 13. is_weekend
## 14. is_holiday_2024
```

```r
write.csv(ev_data, "EV_data_processed.csv", row.names = FALSE)
```

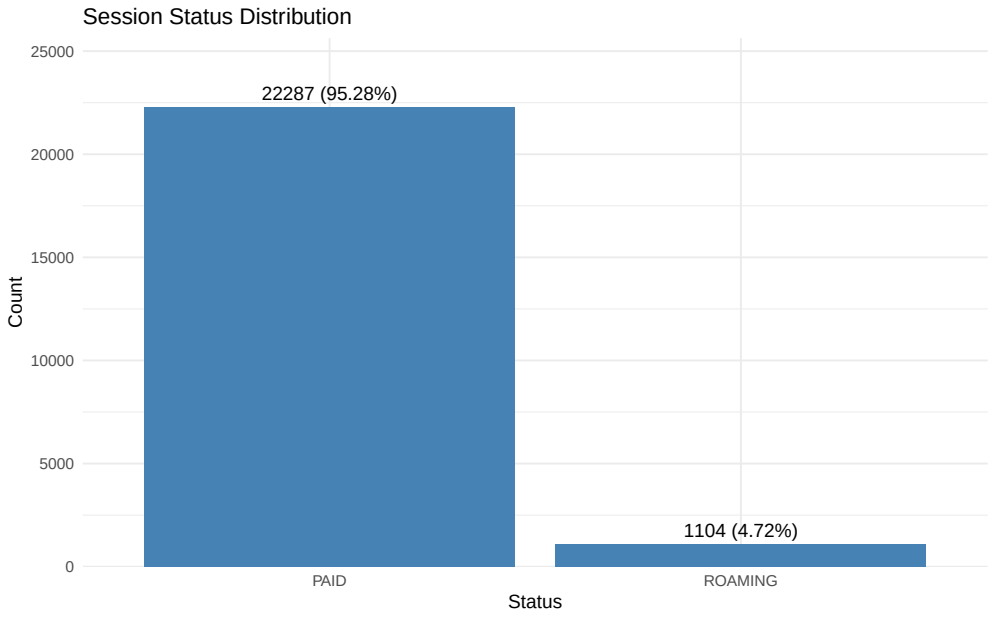
1 Descriptive Analysis

1.0.1 Frequency Tables

1. Frequency table → Session Status distribution; Day of the week distribution; Weekday vs. Weekend distribution; Location distribution; Charging Start Hours distribution; Charging End Hours distribution

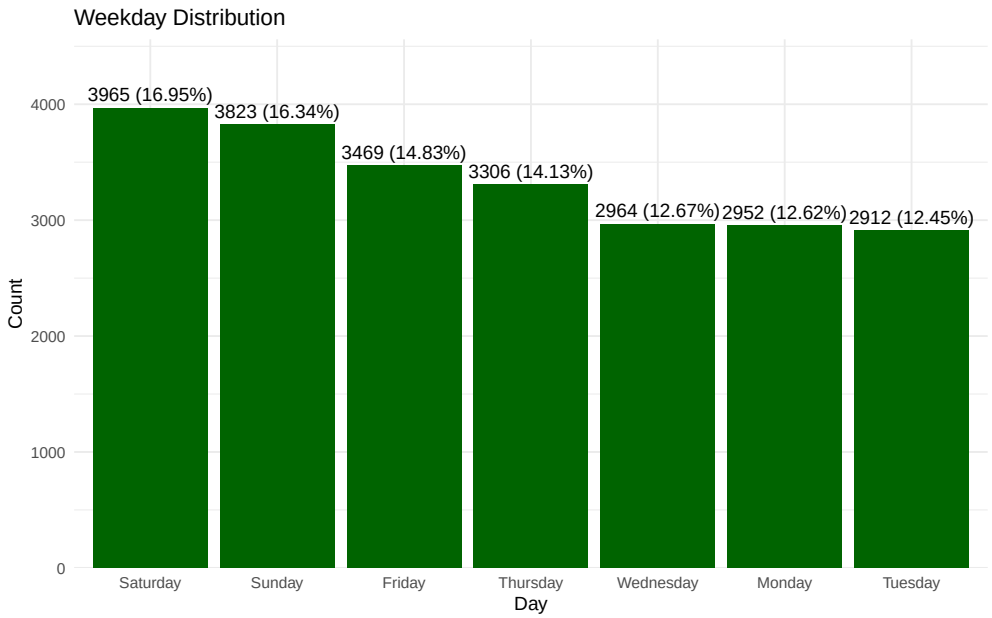
```r
# 1. Session Status distribution
session_status_freq <- ev_data %>%
  count(`Session Status`, sort = TRUE) %>%
  mutate(
    percentage = n / sum(n) * 100,
    percentage = round(percentage, 2)
  )

# Session Status bar chart
ggplot(session_status_freq, aes(x = reorder(`Session Status`, -n), y = n)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = paste0(n, " (", percentage, "%)")), vjust = -0.5) + # Label of
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) + # Additional space to acc
  labs(title = "Session Status Distribution", x = "Status", y = "Count") +
  theme_minimal()
```



```r
# 2. Day of the week distribution
weekday_freq <- ev_data %>%
  count(weekday, sort = TRUE) %>%
  mutate(
    percentage = n / sum(n) * 100,
    percentage = round(percentage, 2)
  )

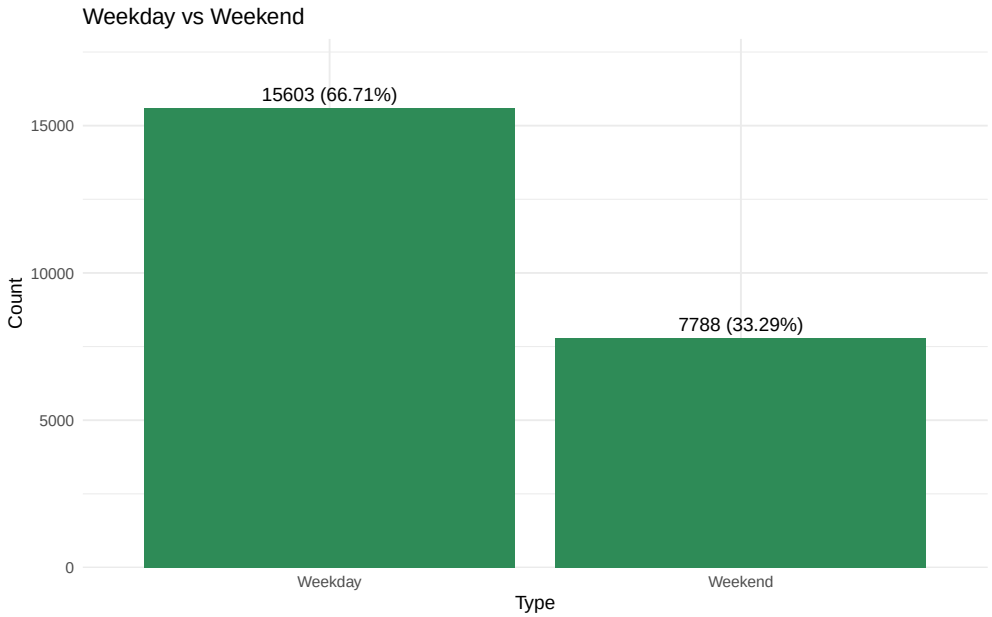
# Day of the week bar chart
ggplot(weekday_freq, aes(x = reorder(weekday, -n), y = n)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  geom_text(aes(label = paste0(n, " (", percentage, "%)")), vjust = -0.5) + # Label of
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) + # Additional space to acc
  labs(title = "Weekday Distribution", x = "Day", y = "Count") +
  theme_minimal()
```




```r
# 3. Weekday vs. Weekend distribution
weekend_freq <- ev_data %>%
  count(is_weekend, sort = TRUE) %>%
  mutate(
    percentage = n / sum(n) * 100,
    percentage = round(percentage, 2),
    type_of_day = ifelse(is_weekend == TRUE, "Weekend", "Weekday")
  )

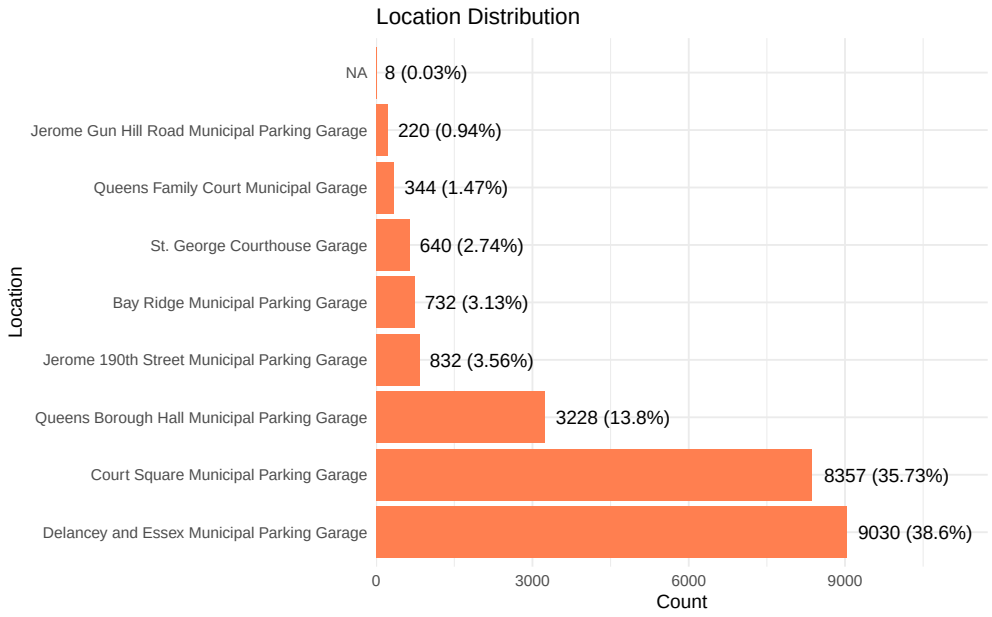
# Weekday vs. Weekend bar chart
```

```r
ggplot(weekend_freq, aes(x = reorder(type_of_day, -n), y = n)) +
  geom_bar(stat = "identity", fill = "seagreen") +
  geom_text(aes(label = paste0(n, " (", percentage, "%)")), vjust = -0.5) + # Label of
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) + # Additional space to acc
  labs(title = "Weekday vs Weekend", x = "Type", y = "Count") +
  theme_minimal()
```




```r
# 4. Location distribution
location_freq <- ev_data %>%
  count(`Location Name`, sort = TRUE) %>%
  mutate(
    percentage = n / sum(n) * 100,
    percentage = round(percentage, 2)
  )

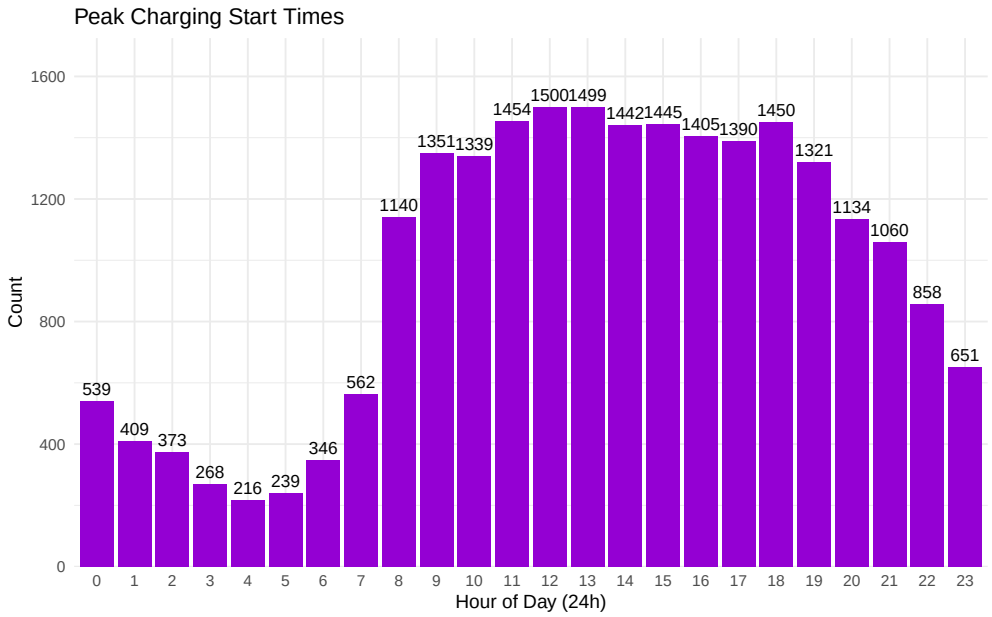
# Location bar chart
ggplot(location_freq, aes(x = reorder(`Location Name`, -n), y = n)) +
  geom_bar(stat = "identity", fill = "coral") +
  coord_flip() +  # Flip for better readability
  geom_text(aes(label = paste0(n, " (", percentage, "%)")), hjust = -0.1) + # Label of
  scale_y_continuous(expand = expansion(mult = c(0, 0.30))) + # Additional space to acc
  labs(title = "Location Distribution", x = "Location", y = "Count") +
  theme_minimal()
```




```r
# 5. Charging Start Hours distribution
hourly_freq <- ev_data %>%
  mutate(
    time_object = hms(`Connected Time`),
    hour_start = hour(time_object)
  ) %>%
  filter(!is.na(hour_start)) %>%
  count(hour_start)

hourly_freq$hour_start <- factor(hourly_freq$hour_start, levels = 0:23)

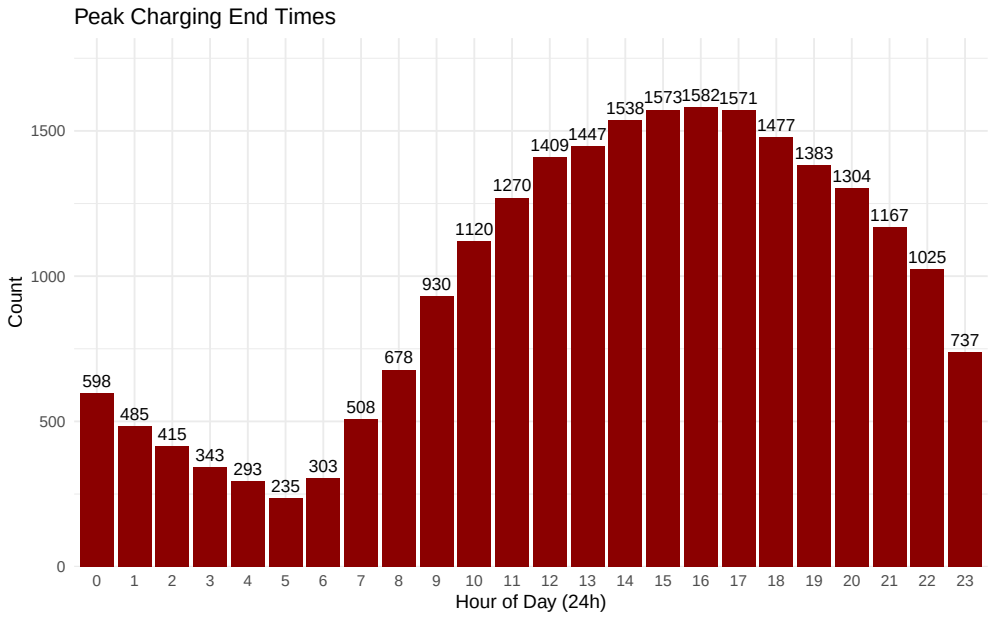
# Charging Start Hours Bar Chart
ggplot(hourly_freq, aes(x = hour_start, y = n)) +
  geom_bar(stat = "identity", fill = "darkviolet") +
  geom_text(aes(label = n), vjust = -0.5, size = 3.5) + # Label of count
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) + # Additional space to acc
  labs(title = "Peak Charging Start Times", x = "Hour of Day (24h)", y = "Count") +
  theme_minimal()
```




```r
# 6. Charging End Hours distribution
end_hourly_freq <- ev_data %>%
  mutate(
    time_object = hms(`Disconnected Time`),
    hour_end = hour(time_object)
  ) %>%
  filter(!is.na(hour_end)) %>%
  count(hour_end)

end_hourly_freq$hour_end <- factor(end_hourly_freq$hour_end, levels = 0:23)

# Charging End Hours Bar Chart
ggplot(end_hourly_freq, aes(x = hour_end, y = n)) +
  geom_bar(stat = "identity", fill = "darkred") +
  geom_text(aes(label = n), vjust = -0.5, size = 3.5) + # Label of count
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) + # Additional space to acc
  labs(title = "Peak Charging End Times", x = "Hour of Day (24h)", y = "Count") +
  theme_minimal()
```




```r
### Center and Dispersion Statistics

# Function to calculate mode (R doesn't have a built-in mode function)
get_mode <- function(x) {
  x <- na.omit(x)
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}

# Function to calculate all statistics for a variable
calc_stats <- function(x, var_name) {
  x <- na.omit(x)

  data.frame(
    Variable = var_name,
    Mean = round(mean(x), 3),
    Median = round(median(x), 3),
    Mode = round(get_mode(x), 3),
    Trimmed_Mean_10pct = round(mean(x, trim = 0.10), 3),
    IQR = round(IQR(x), 3),
    Variance = round(var(x), 3),
    Std_Dev = round(sd(x), 3),
    CV = round((sd(x) / mean(x)) * 100, 2),  # Coefficient of Variation as percentage
    Skewness = round(skewness(x), 3)
  )
}
```

```r
# Calculate statistics for Energy Provided and Charge Duration
energy_stats <- calc_stats(ev_data$`Energy Provided (kWh)`, "Energy Provided (kWh)")
duration_stats <- calc_stats(ev_data$`Charge Duration (min)`, "Charge Duration (min)")

# Combine into one table
summary_stats <- rbind(energy_stats, duration_stats)

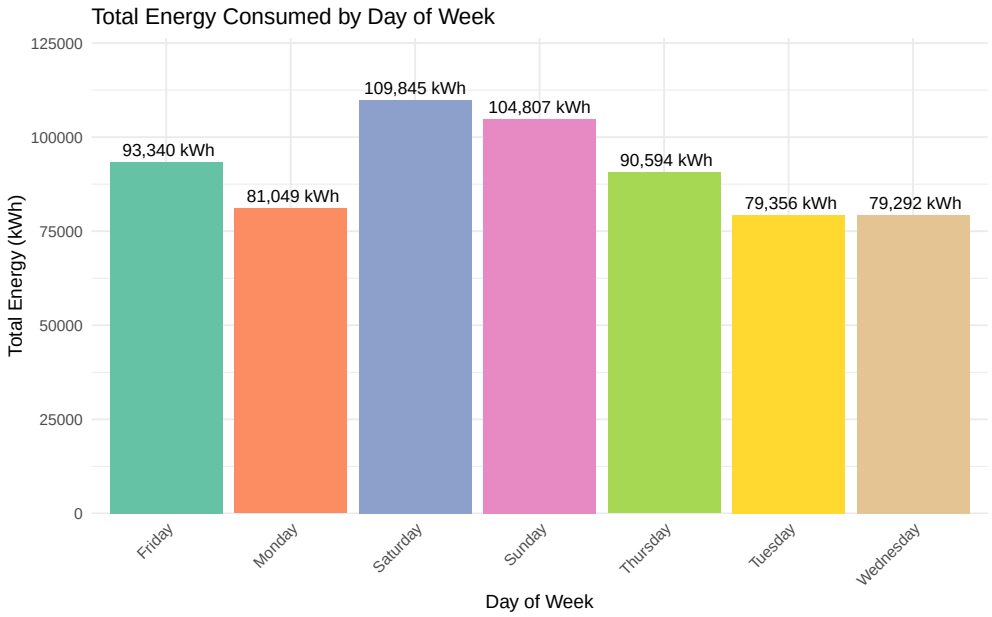
# Display the table
kable(summary_stats,
      caption = "Summary Statistics: Measures of Center and Dispersion",
      col.names = c("Variable", "Mean", "Median", "Mode", "Trimmed Mean (10%)",
                    "IQR", "Variance", "Std Dev", "CV (%)", "Skewness"),
      align = "lrrrrrrrrr") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE,
                position = "center")
```

Table 1: Summary Statistics: Measures of Center and Dispersion

Variable	Mean	Median	Mode	Trimmed Mean (10%)	IQR	Variance	Std D
Energy Provided (kWh)	27.288	24.449	0.008	25.489	30.144	428.636	20.7
Charge Duration (min)	130.285	58.083	8.000	78.612	82.575	68687.433	262.0

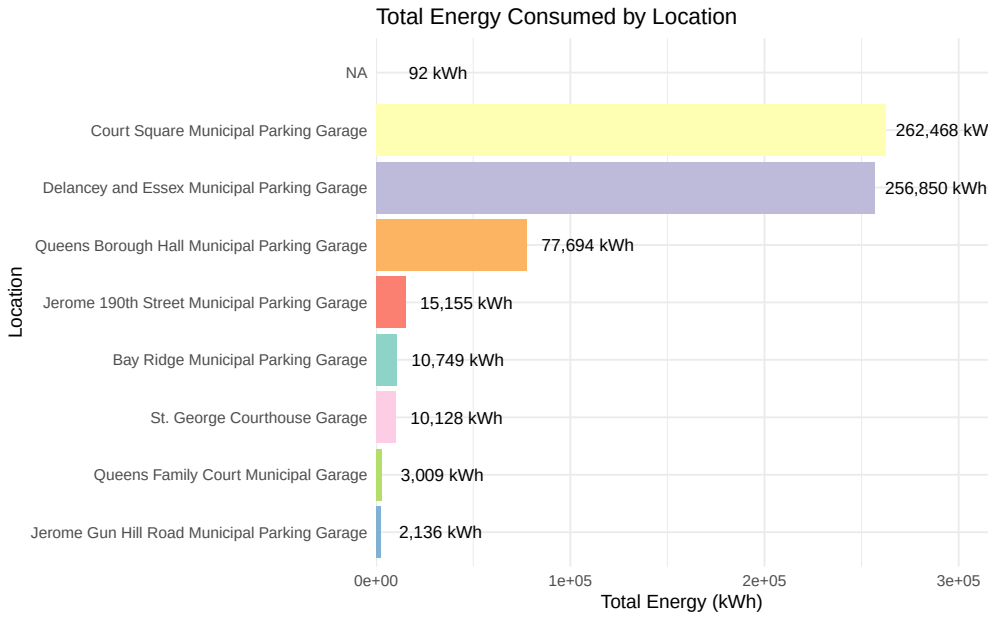
```r
# Total Energy Consumed by Day of Week
energy_by_day <- ev_data %>%
    group_by(weekday) %>%
    summarise(total_energy = sum(`Energy Provided (kWh)`, na.rm = TRUE),
        num_sessions = n(), avg_energy = mean(`Energy Provided (kWh)`,
            na.rm = TRUE)) %>%
    mutate(total_energy_rounded = round(total_energy, 0), avg_energy_rounded = round(avg
        2))

# Bar plot of Total Energy by Weekday
ggplot(energy_by_day, aes(x = weekday, y = total_energy, fill = weekday)) +
    geom_bar(stat = "identity") + geom_text(aes(label = paste0(format(total_energy_roun
    big.mark = ","), " kWh")), vjust = -0.5, size = 3.5) + scale_y_continuous(expand =
    0.15))) + labs(title = "Total Energy Consumed by Day of Week",
    x = "Day of Week", y = "Total Energy (kWh)") + theme_minimal() +
    theme(legend.position = "none", axis.text.x = element_text(angle = 45,
        hjust = 1)) + scale_fill_brewer(palette = "Set2")
```

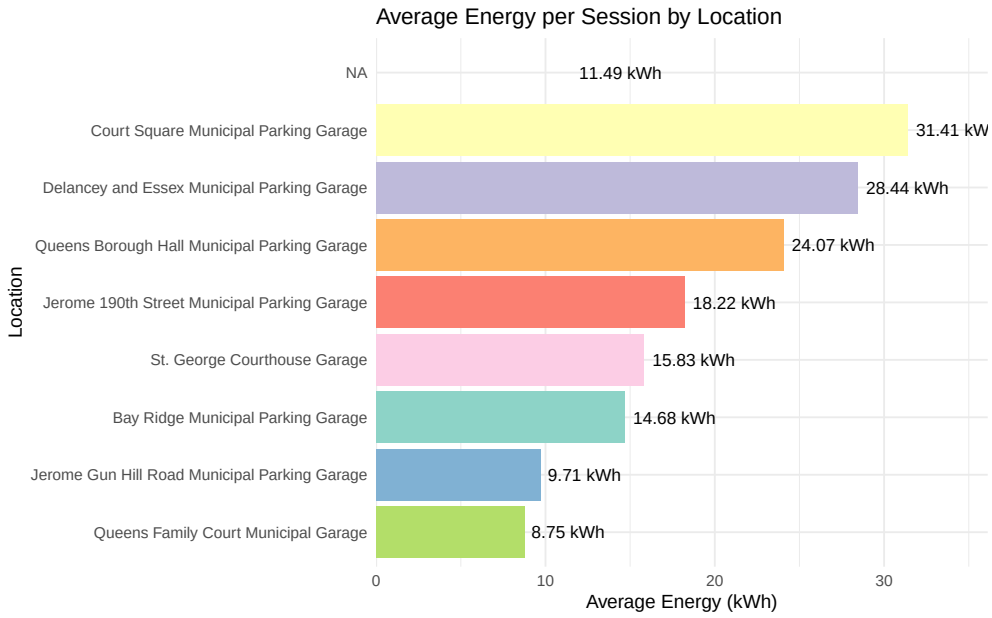



```
# Total Energy Consumed by Location
energy_by_location <- ev_data %>%
    group_by(`Location Name`) %>%
    summarise(total_energy = sum(`Energy Provided (kWh)`, na.rm = TRUE),
        num_sessions = n(), avg_energy = mean(`Energy Provided (kWh)`,
            na.rm = TRUE)) %>%
    mutate(total_energy_rounded = round(total_energy, 0), avg_energy_rounded = round(avg
        2))

# Bar plot of Total Energy by Location
ggplot(energy_by_location, aes(x = reorder(`Location Name`, total_energy),
    y = total_energy, fill = `Location Name`)) + geom_bar(stat = "identity") +
    geom_text(aes(label = paste0(format(total_energy_rounded,
        big.mark = ","), " kWh")), hjust = -0.1, size = 3.5) +
    coord_flip() + scale_y_continuous(expand = expansion(mult = c(0,
    0.2))) + labs(title = "Total Energy Consumed by Location",
    x = "Location", y = "Total Energy (kWh)") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_brewer(palette = "Set3")
```




```
# Bar plot of Average Energy per Session by Location
ggplot(energy_by_location, aes(x = reorder(`Location Name`, avg_energy),
    y = avg_energy, fill = `Location Name`)) + geom_bar(stat = "identity") +
    geom_text(aes(label = paste0(avg_energy_rounded, " kWh")),
        hjust = -0.1, size = 3.5) + coord_flip() + scale_y_continuous(expand = expansio
    0.15))) + labs(title = "Average Energy per Session by Location",
    x = "Location", y = "Average Energy (kWh)") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_brewer(palette = "Set3")
```

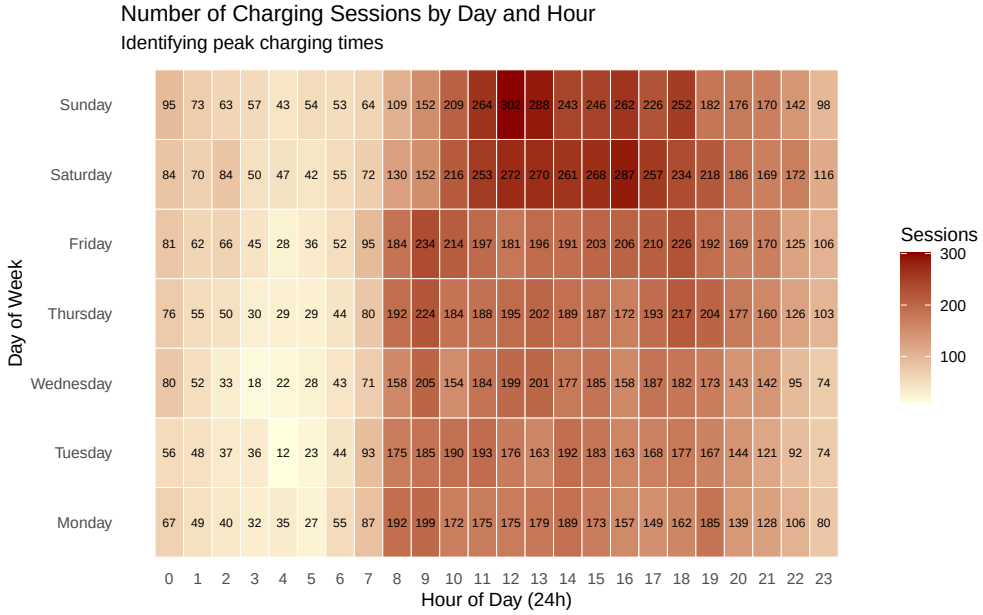


```r
### Heatmap of Temporal Usage (Weekday × Hour of Day)

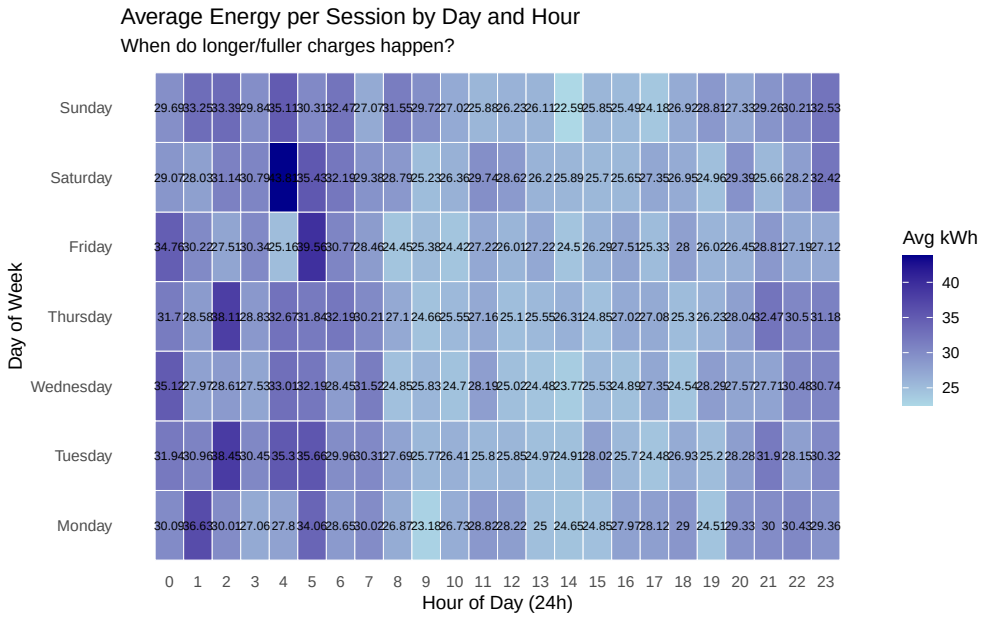
# Prepare data: extract hour and summarize
temporal_data <- ev_data %>%
    mutate(time_object = hms(`Connected Time`), hour = hour(time_object)) %>%
    filter(!is.na(hour)) %>%
    group_by(weekday, hour) %>%
    summarise(num_sessions = n(), avg_energy = round(mean(`Energy Provided (kWh)`,
        na.rm = TRUE), 2), total_energy = round(sum(`Energy Provided (kWh)`,
        na.rm = TRUE), 0), .groups = "drop")

# Ensure proper weekday order
temporal_data$weekday <- factor(temporal_data$weekday, levels = c("Monday",
    "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",
    "Sunday"))

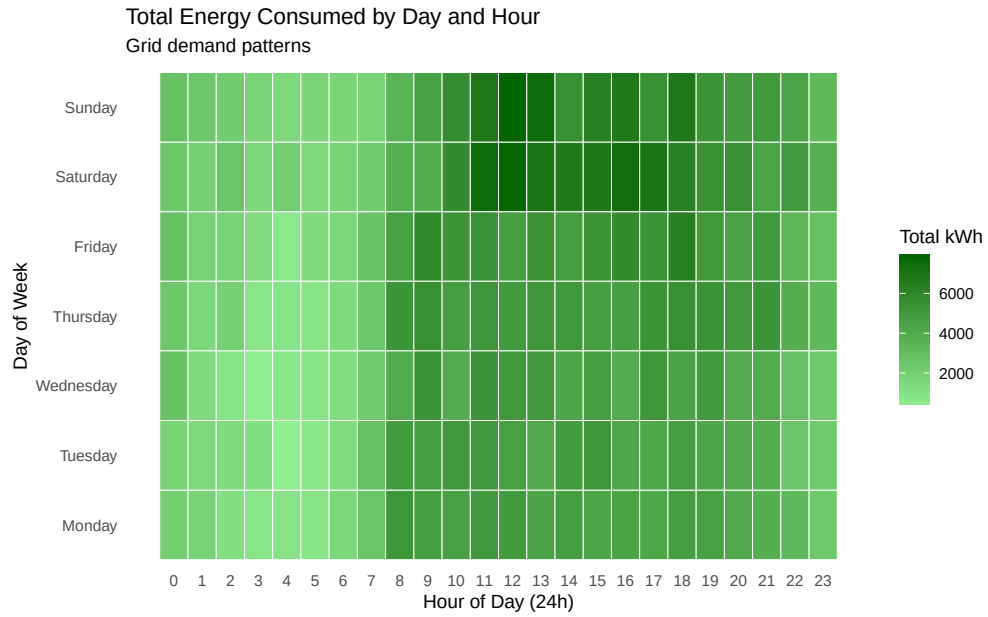
# 1. Heatmap: Number of Sessions by Weekday × Hour
ggplot(temporal_data, aes(x = hour, y = weekday, fill = num_sessions)) +
    geom_tile(color = "white") + geom_text(aes(label = num_sessions),
    size = 2.5, color = "black") + scale_fill_gradient(low = "lightyellow",
    high = "darkred") + scale_x_continuous(breaks = 0:23) + labs(title = "Number of Cha
    subtitle = "Identifying peak charging times", x = "Hour of Day (24h)",
    y = "Day of Week", fill = "Sessions") + theme_minimal() +
    theme(panel.grid = element_blank())
```



```r
# 2. Heatmap: Average Energy per Session by Weekday × Hour
ggplot(temporal_data, aes(x = hour, y = weekday, fill = avg_energy)) +
    geom_tile(color = "white") + geom_text(aes(label = avg_energy),
    size = 2.5, color = "black") + scale_fill_gradient(low = "lightblue",
    high = "darkblue") + scale_x_continuous(breaks = 0:23) +
    labs(title = "Average Energy per Session by Day and Hour",
        subtitle = "When do longer/fuller charges happen?", x = "Hour of Day (24h)",
        y = "Day of Week", fill = "Avg kWh") + theme_minimal() +
    theme(panel.grid = element_blank())
```




```r
# 3. Heatmap: Total Energy by Weekday × Hour
ggplot(temporal_data, aes(x = hour, y = weekday, fill = total_energy)) +
    geom_tile(color = "white") + scale_fill_gradient(low = "lightgreen",
    high = "darkgreen") + scale_x_continuous(breaks = 0:23) +
    labs(title = "Total Energy Consumed by Day and Hour", subtitle = "Grid demand patter
        x = "Hour of Day (24h)", y = "Day of Week", fill = "Total kWh") +
    theme_minimal() + theme(panel.grid = element_blank())
```



#Inferencial Statistics:

1.0.2 Weekend vs weekday differences

- Two-sample t-tests (or Wilcoxon based on normality) on Charge Duration and Energy Provided:
- H0: Mean of Weekend = Mean of Weekdays.
- 95% CI for the difference in means.
- Have to check for normality (or lack of it) using the histograms + Q–Q plots from desc stats.

```
### Weekend vs Weekday Differences

# Separate data into weekend and weekday groups
weekday_data <- ev_data %>%
    filter(is_weekend == FALSE)
weekend_data <- ev_data %>%
    filter(is_weekend == TRUE)

# Display sample sizes
cat("Sample Sizes:\n")
```

```
## Sample Sizes:
```

```r
cat("Weekday sessions:", nrow(weekday_data), "\n")
```

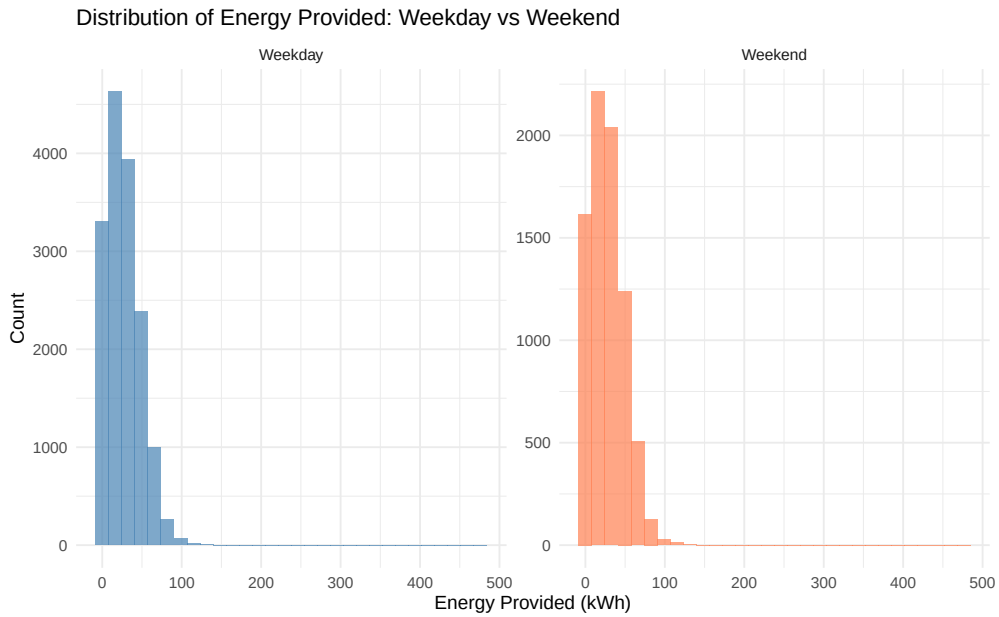
```
## Weekday sessions: 15603
```

```r
cat("Weekend sessions:", nrow(weekend_data), "\n")
```

```
## Weekend sessions: 7788
```

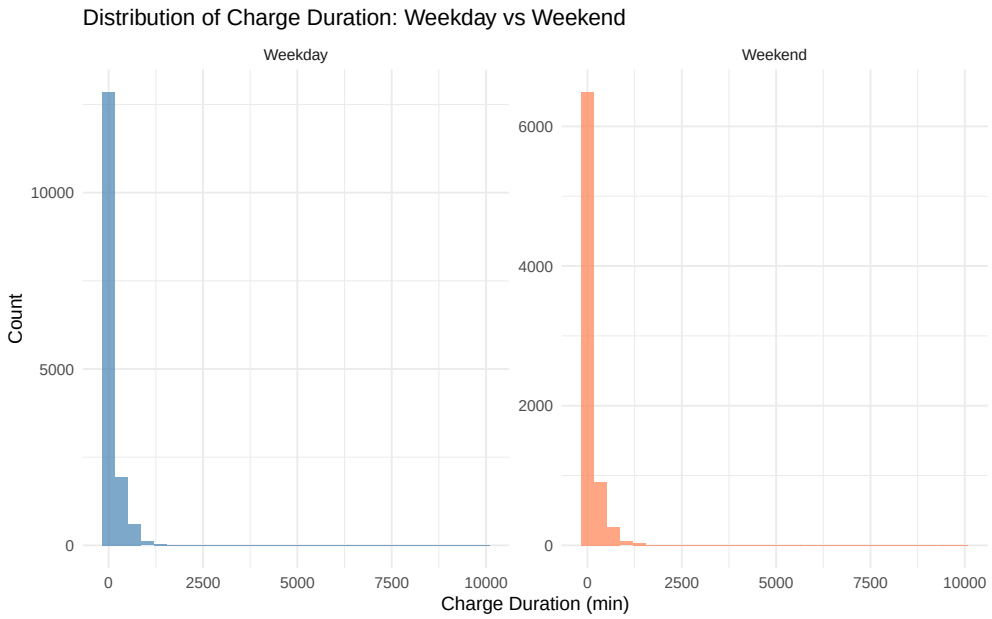
```r
#----------------------------------------
# 1. Normality Check: Histograms
#----------------------------------------

# Histogram: Energy Provided by Weekend vs Weekday
ggplot(ev_data, aes(x = `Energy Provided (kWh)`, fill = is_weekend)) +
    geom_histogram(bins = 30, alpha = 0.7, position = "identity") +
    facet_wrap(~ifelse(is_weekend, "Weekend", "Weekday"), scales = "free_y") +
    labs(title = "Distribution of Energy Provided: Weekday vs Weekend",
        x = "Energy Provided (kWh)", y = "Count") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_manual(values = c("steelblue",
    "coral"))
```



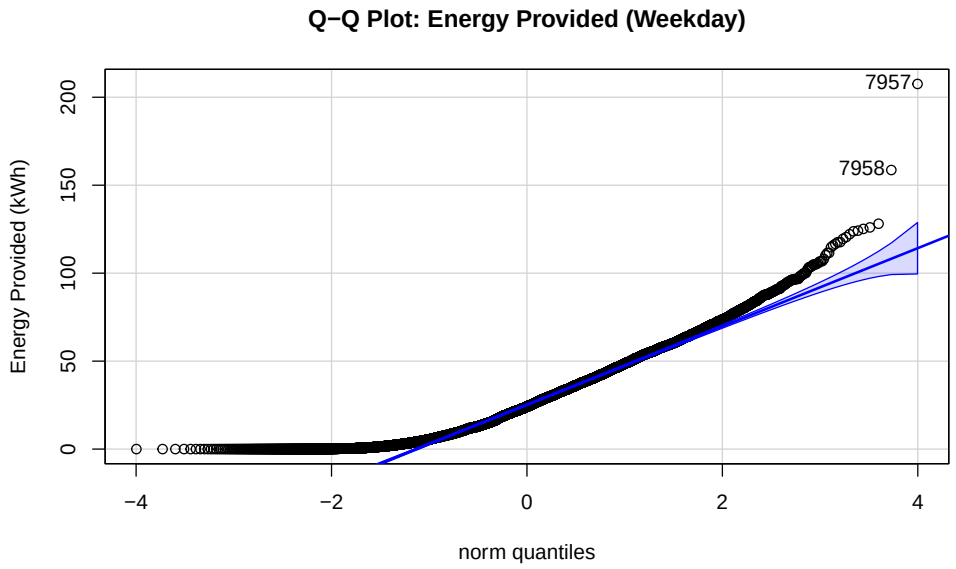

```r
# Histogram: Charge Duration by Weekend vs Weekday
ggplot(ev_data, aes(x = `Charge Duration (min)`, fill = is_weekend)) +
    geom_histogram(bins = 30, alpha = 0.7, position = "identity") +
    facet_wrap(~ifelse(is_weekend, "Weekend", "Weekday"), scales = "free_y") +
```

```
    labs(title = "Distribution of Charge Duration: Weekday vs Weekend",
        x = "Charge Duration (min)", y = "Count") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_manual(values = c("steelblue",
    "coral"))
```



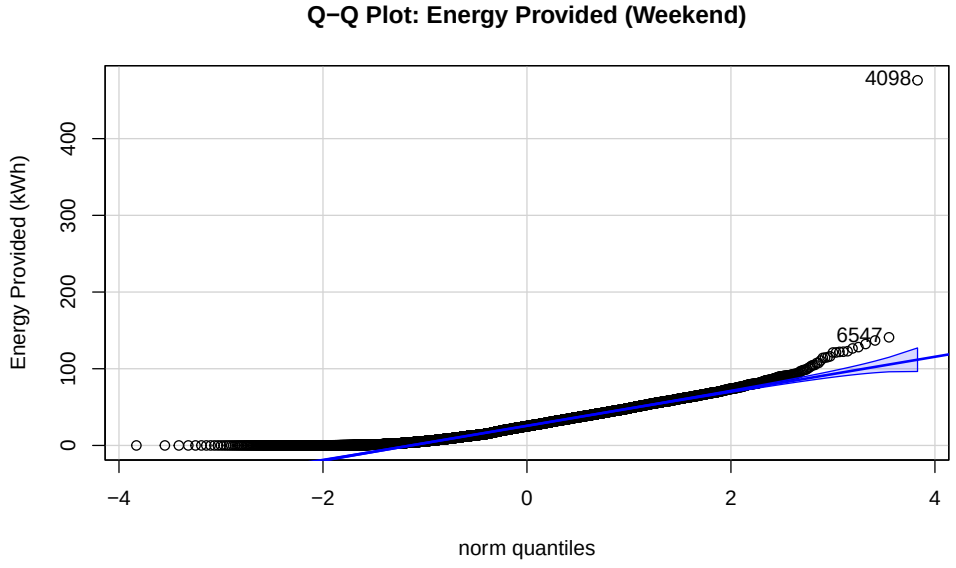

```
#----------------------------------------
# 2. Normality Check: Q-Q Plots
#----------------------------------------

# Q-Q Plot: Energy Provided - Weekday
qqPlot(weekday_data$`Energy Provided (kWh)`, main = "Q-Q Plot: Energy Provided (Weekday)
    ylab = "Energy Provided (kWh)")
```



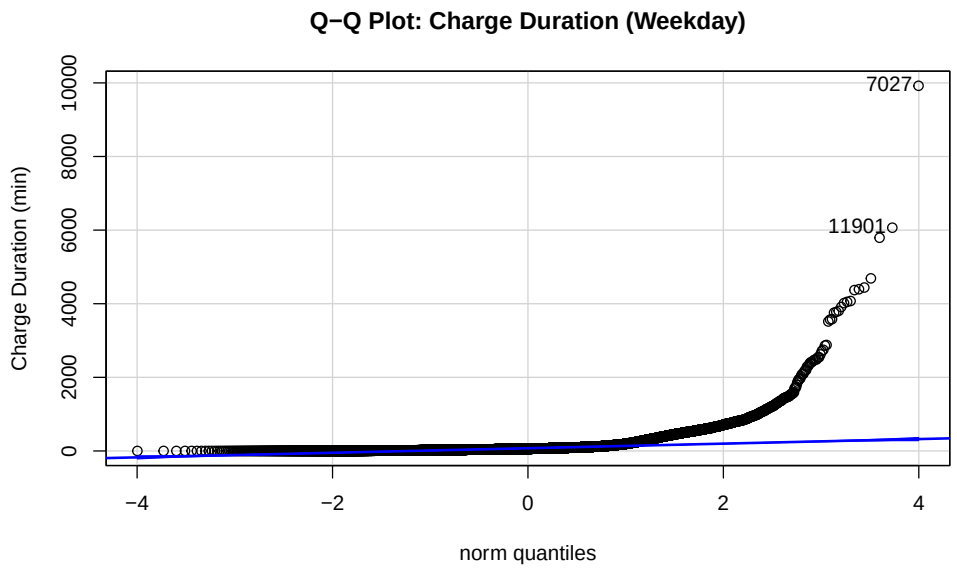

```
## [1] 7957 7958
```

```
# Q-Q Plot: Energy Provided - Weekend
qqPlot(weekend_data$`Energy Provided (kWh)`, main = "Q-Q Plot: Energy Provided (Weekend)
    ylab = "Energy Provided (kWh)")
```



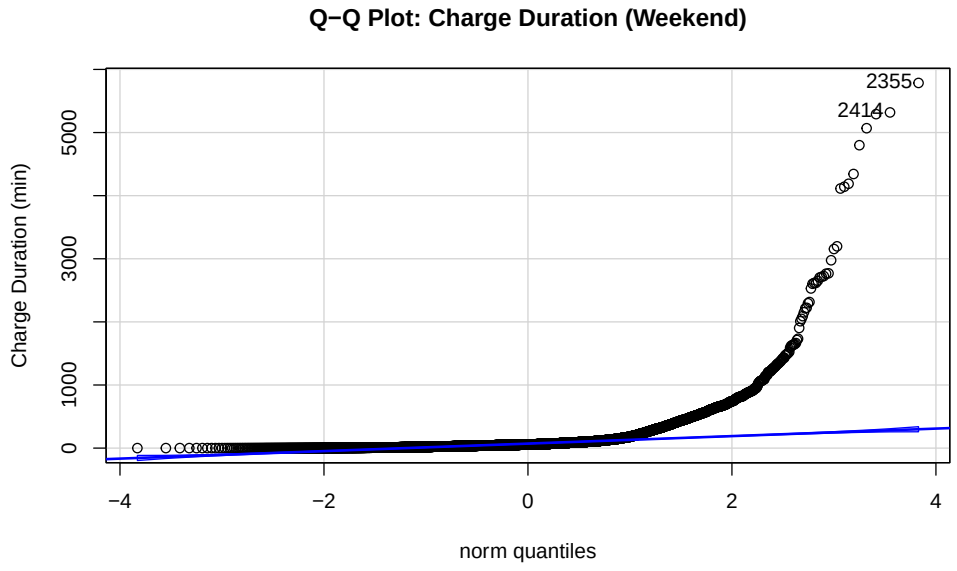

```
## [1] 4098 6547
```

```r
# Q-Q Plot: Charge Duration - Weekday
qqPlot(weekday_data$`Charge Duration (min)`, main = "Q-Q Plot: Charge Duration (Weekday)",
    ylab = "Charge Duration (min)")
```




```
## [1]  7027 11901
```

```r
# Q-Q Plot: Charge Duration - Weekend
qqPlot(weekend_data$`Charge Duration (min)`, main = "Q-Q Plot: Charge Duration (Weekend)",
    ylab = "Charge Duration (min)")
```



```
## [1] 2355 2414
```

```
#----------------------------------------
# 3. Formal Normality Test (Shapiro-Wilk on sample)
#----------------------------------------
# Note: Shapiro-Wilk limited to 5000 observations, so we
# sample

set.seed(123)
shapiro_weekday_energy <- shapiro.test(sample(weekday_data$`Energy Provided (kWh)`,
    5000))
shapiro_weekend_energy <- shapiro.test(sample(weekend_data$`Energy Provided (kWh)`,
    min(5000, nrow(weekend_data))))
shapiro_weekday_duration <- shapiro.test(sample(weekday_data$`Charge Duration (min)`,
    5000))
shapiro_weekend_duration <- shapiro.test(sample(weekend_data$`Charge Duration (min)`,
    min(5000, nrow(weekend_data))))

cat("\n--- Shapiro-Wilk Normality Tests ---\n")
```

```
##
## --- Shapiro-Wilk Normality Tests ---
```

```
cat("Energy (Weekday): W =", round(shapiro_weekday_energy$statistic,
    4), ", p-value =", format(shapiro_weekday_energy$p.value,
    scientific = TRUE), "\n")
```

```
## Energy (Weekday): W = 0.9451 , p-value = 1.155929e-39
```

```
cat("Energy (Weekend): W =", round(shapiro_weekend_energy$statistic,
    4), ", p-value =", format(shapiro_weekend_energy$p.value,
    scientific = TRUE), "\n")
```

```
## Energy (Weekend): W = 0.8901 , p-value = 8.185068e-51
```

```
cat("Duration (Weekday): W =", round(shapiro_weekday_duration$statistic,
    4), ", p-value =", format(shapiro_weekday_duration$p.value,
    scientific = TRUE), "\n")
```

```
## Duration (Weekday): W = 0.4702 , p-value = 2.087111e-81
```

```r
cat("Duration (Weekend): W =", round(shapiro_weekend_duration$statistic,
    4), ", p-value =", format(shapiro_weekend_duration$p.value,
    scientific = TRUE), "\n")
```

```
## Duration (Weekend): W = 0.383 , p-value = 9.244873e-85
```

```r
#----------------------------------------
# 4. Hypothesis Tests
#----------------------------------------
# Based on normality results, run both t-test and Wilcoxon
# for comparison H0: Mean(Weekend) = Mean(Weekday) Ha:
# Mean(Weekend)  Mean(Weekday) Significance level:  =
# 0.05

cat("\n===== ENERGY PROVIDED: Weekend vs Weekday =====\n")
```

```
##
## ===== ENERGY PROVIDED: Weekend vs Weekday =====
```

```r
# Descriptive comparison
cat("\nDescriptive Statistics:\n")
```

```
##
## Descriptive Statistics:
```

```r
cat("Weekday - Mean:", round(mean(weekday_data$`Energy Provided (kWh)`),
    3), "kWh, Median:", round(median(weekday_data$`Energy Provided (kWh)`),
    3), "kWh\n")
```

```
## Weekday - Mean: 27.151 kWh, Median: 24.106 kWh
```

```r
cat("Weekend - Mean:", round(mean(weekend_data$`Energy Provided (kWh)`),
    3), "kWh, Median:", round(median(weekend_data$`Energy Provided (kWh)`),
    3), "kWh\n")
```

```
## Weekend - Mean: 27.562 kWh, Median: 25.14 kWh
```

```r
# Two-sample t-test (Welch's t-test, does not assume equal
# variances)
t_test_energy <- t.test(`Energy Provided (kWh)` ~ is_weekend,
    data = ev_data, conf.level = 0.95)
cat("\n--- Two-Sample t-Test (Welch) ---\n")
```

```
##
## --- Two-Sample t-Test (Welch) ---
```

```
cat("t-statistic:", round(t_test_energy$statistic, 4), "\n")
```

```
## t-statistic: -1.4153
```

```
cat("Degrees of freedom:", round(t_test_energy$parameter, 2),
    "\n")
```

```
## Degrees of freedom: 15099.4
```

```
cat("p-value:", format(t_test_energy$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 1.570139e-01
```

```
cat("95% CI for difference (Weekday - Weekend):", round(t_test_energy$conf.int[1],
    3), "to", round(t_test_energy$conf.int[2], 3), "kWh\n")
```

```
## 95% CI for difference (Weekday - Weekend): -0.981 to 0.158 kWh
```

```
# Wilcoxon rank-sum test (non-parametric alternative)
wilcox_energy <- wilcox.test(`Energy Provided (kWh)` ~ is_weekend,
    data = ev_data, conf.int = TRUE)
cat("\n--- Wilcoxon Rank-Sum Test ---\n")
```

```
##
## --- Wilcoxon Rank-Sum Test ---
```

```
cat("W statistic:", wilcox_energy$statistic, "\n")
```

```
## W statistic: 60137316
```

```
cat("p-value:", format(wilcox_energy$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 2.021458e-01
```

```r
cat("95% CI for difference:", round(wilcox_energy$conf.int[1],
    3), "to", round(wilcox_energy$conf.int[2], 3), "kWh\n")
```

```
## 95% CI for difference: -0.78 to 0.149 kWh
```

```r
cat("\n===== CHARGE DURATION: Weekend vs Weekday =====\n")
```

```
##
## ===== CHARGE DURATION: Weekend vs Weekday =====
```

```r
# Descriptive comparison
cat("\nDescriptive Statistics:\n")
```

```
##
## Descriptive Statistics:
```

```r
cat("Weekday - Mean:", round(mean(weekday_data$`Charge Duration (min)`),
    3), "min, Median:", round(median(weekday_data$`Charge Duration (min)`),
    3), "min\n")
```

```
## Weekday - Mean: 130.035 min, Median: 58.917 min
```

```r
cat("Weekend - Mean:", round(mean(weekend_data$`Charge Duration (min)`),
    3), "min, Median:", round(median(weekend_data$`Charge Duration (min)`),
    3), "min\n")
```

```
## Weekend - Mean: 130.786 min, Median: 56.747 min
```

```r
# Two-sample t-test
t_test_duration <- t.test(`Charge Duration (min)` ~ is_weekend,
    data = ev_data, conf.level = 0.95)
cat("\n--- Two-Sample t-Test (Welch) ---\n")
```

```
##
## --- Two-Sample t-Test (Welch) ---
```

```r
cat("t-statistic:", round(t_test_duration$statistic, 4), "\n")
```

```
## t-statistic: -0.2013
```

```r
cat("Degrees of freedom:", round(t_test_duration$parameter, 2),
    "\n")
```

```
## Degrees of freedom: 14563.67
```

```r
cat("p-value:", format(t_test_duration$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 8.405032e-01
```

```r
cat("95% CI for difference (Weekday - Weekend):", round(t_test_duration$conf.int[1],
    3), "to", round(t_test_duration$conf.int[2], 3), "min\n")
```

```
## 95% CI for difference (Weekday - Weekend): -8.06 to 6.559 min
```

```r
# Wilcoxon rank-sum test
wilcox_duration <- wilcox.test(`Charge Duration (min)` ~ is_weekend,
    data = ev_data, conf.int = TRUE)
cat("\n--- Wilcoxon Rank-Sum Test ---\n")
```

```
##
## --- Wilcoxon Rank-Sum Test ---
```

```r
cat("W statistic:", wilcox_duration$statistic, "\n")
```

```
## W statistic: 61787124
```

```r
cat("p-value:", format(wilcox_duration$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 3.448716e-02
```

```r
cat("95% CI for difference:", round(wilcox_duration$conf.int[1],
    3), "to", round(wilcox_duration$conf.int[2], 3), "min\n")
```

```
## 95% CI for difference: 0.1 to 2.85 min
```

```r
#----------------------------------------
# 5. Summary Table
#----------------------------------------
results_summary <- data.frame(Variable = c("Energy Provided (kWh)",
    "Charge Duration (min)"), Weekday_Mean = c(round(mean(weekday_data$`Energy Provided
    2), round(mean(weekday_data$`Charge Duration (min)`), 2)),
    Weekend_Mean = c(round(mean(weekend_data$`Energy Provided (kWh)`),
        2), round(mean(weekend_data$`Charge Duration (min)`),
        2)), t_statistic = c(round(t_test_energy$statistic, 3),
        round(t_test_duration$statistic, 3)), p_value = c(format(t_test_energy$p.value,
        scientific = TRUE), format(t_test_duration$p.value, scientific = TRUE)),
    CI_lower = c(round(t_test_energy$conf.int[1], 3), round(t_test_duration$conf.int[1],
        3)), CI_upper = c(round(t_test_energy$conf.int[2], 3),
        round(t_test_duration$conf.int[2], 3)), Significant = c(ifelse(t_test_energy$p.v
        0.05, "Yes", "No"), ifelse(t_test_duration$p.value <
        0.05, "Yes", "No")))

kable(results_summary, caption = "Weekend vs Weekday Comparison: Two-Sample t-Test Resul
    col.names = c("Variable", "Weekday Mean", "Weekend Mean",
        "t-statistic", "p-value", "95% CI Lower", "95% CI Upper",
        "Significant?"), align = "lrrrrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 2: Weekend vs Weekday Comparison: Two-Sample t-Test Results (= 0.05)

Variable	Weekday Mean	Weekend Mean	t-statistic	p-value	95% CI Lower
Energy Provided (kWh)	27.15	27.56	-1.415	1.570139e-01	-0.981
Charge Duration (min)	130.04	130.79	-0.201	8.405032e-01	-8.060

```r
### Weekend vs Weekday Differences Significance Level:  =
### 0.05

alpha <- 0.05

# Separate data into weekend and weekday groups
weekday_data <- ev_data %>%
    filter(is_weekend == FALSE)
weekend_data <- ev_data %>%
    filter(is_weekend == TRUE)
```

```r
# Display sample sizes
cat("Sample Sizes:\n")
```

```
## Sample Sizes:
```

```r
cat("Weekday sessions:", nrow(weekday_data), "\n")
```

```
## Weekday sessions: 15603
```

```r
cat("Weekend sessions:", nrow(weekend_data), "\n")
```

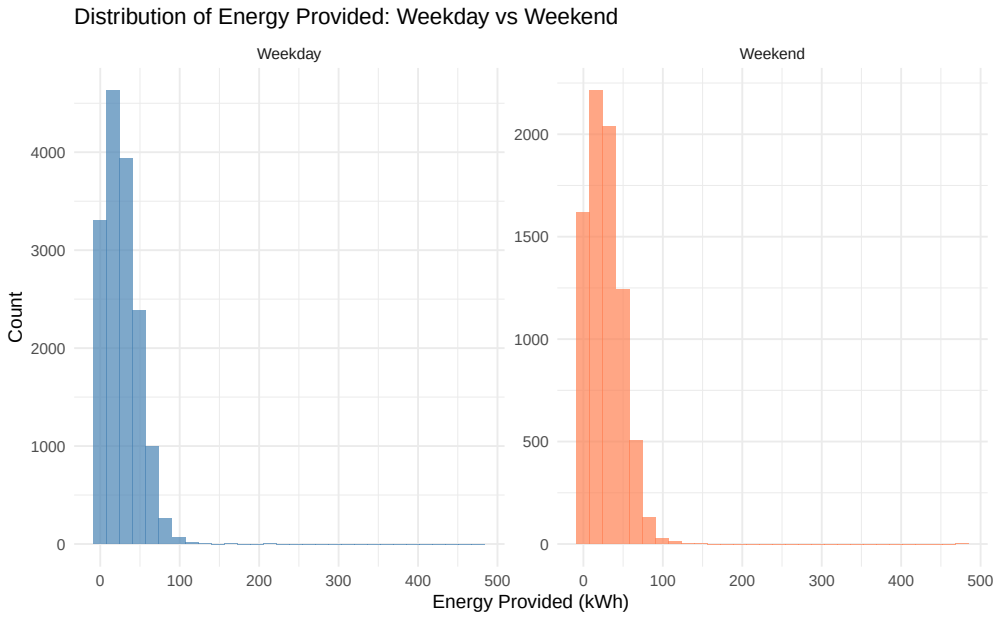
```
## Weekend sessions: 7788
```

```r
cat("Significance Level:   =", alpha, "\n")
```

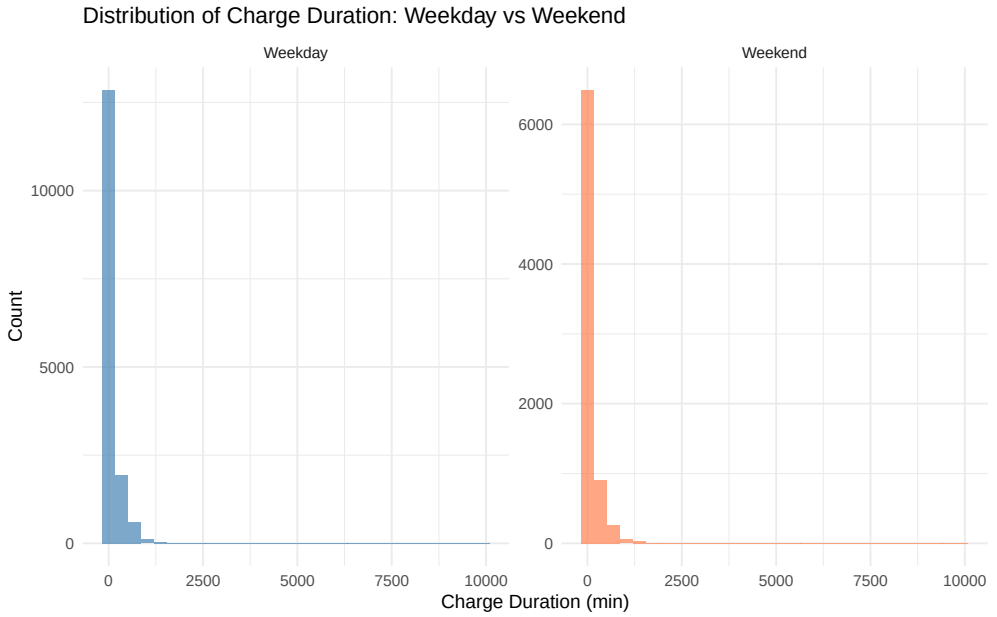
```
## Significance Level:   = 0.05
```

```r
#----------------------------------------
# 1. Normality Check: Histograms
#----------------------------------------

# Histogram: Energy Provided by Weekend vs Weekday
ggplot(ev_data, aes(x = `Energy Provided (kWh)`, fill = is_weekend)) +
    geom_histogram(bins = 30, alpha = 0.7, position = "identity") +
    facet_wrap(~ifelse(is_weekend, "Weekend", "Weekday"), scales = "free_y") +
    labs(title = "Distribution of Energy Provided: Weekday vs Weekend",
        x = "Energy Provided (kWh)", y = "Count") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_manual(values = c("steelblue",
    "coral"))
```

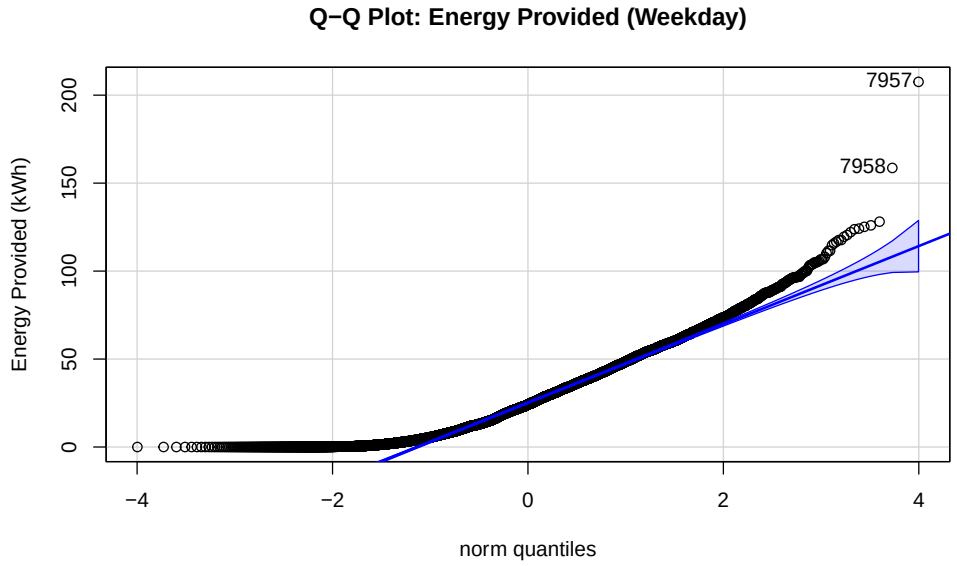



```
# Histogram: Charge Duration by Weekend vs Weekday
ggplot(ev_data, aes(x = `Charge Duration (min)`, fill = is_weekend)) +
    geom_histogram(bins = 30, alpha = 0.7, position = "identity") +
    facet_wrap(~ifelse(is_weekend, "Weekend", "Weekday"), scales = "free_y") +
    labs(title = "Distribution of Charge Duration: Weekday vs Weekend",
        x = "Charge Duration (min)", y = "Count") + theme_minimal() +
    theme(legend.position = "none") + scale_fill_manual(values = c("steelblue",
    "coral"))
```



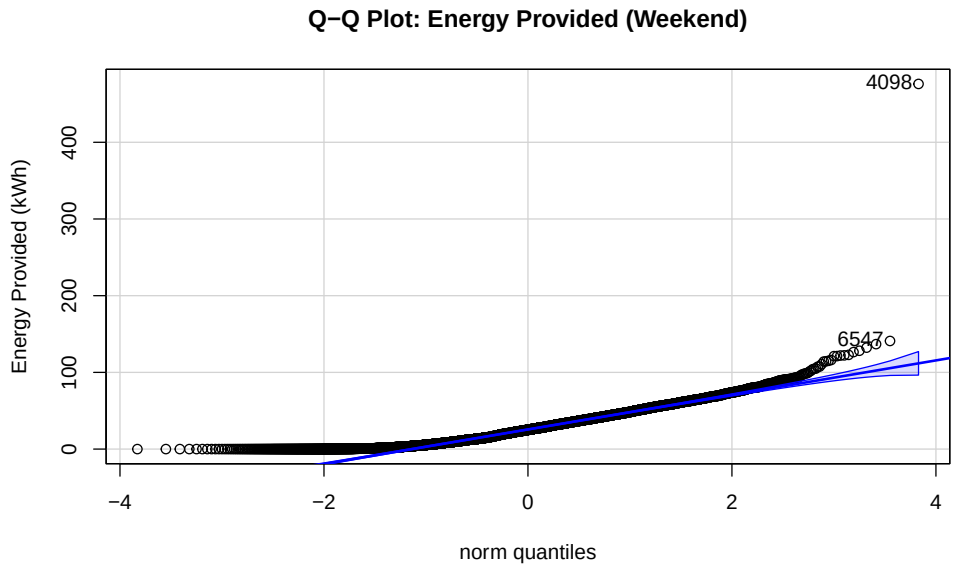
```r
#----------------------------------------
# 2. Normality Check: Q-Q Plots
#----------------------------------------

# Q-Q Plot: Energy Provided - Weekday
qqPlot(weekday_data$`Energy Provided (kWh)`, main = "Q-Q Plot: Energy Provided (Weekday)
    ylab = "Energy Provided (kWh)")
```



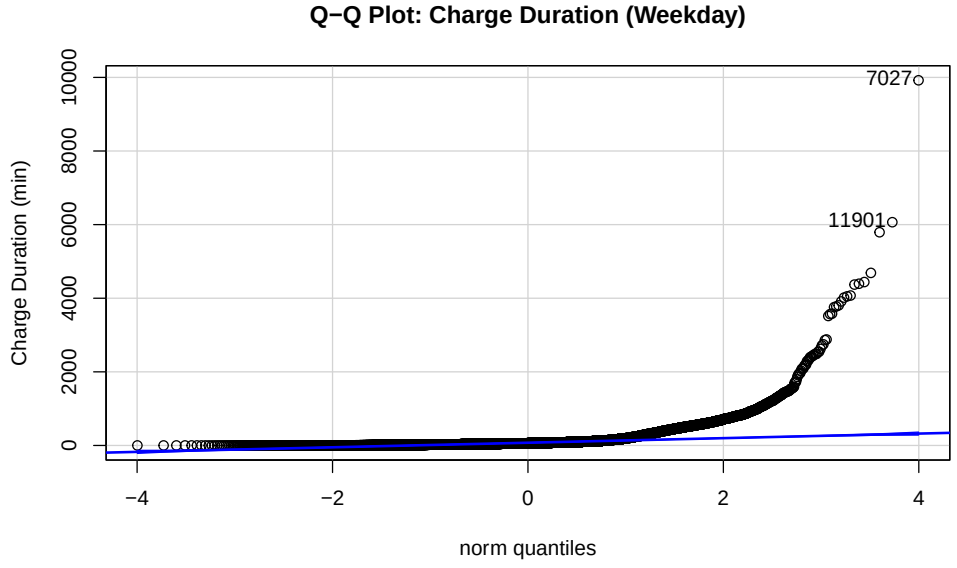

```
## [1] 7957 7958
```

```r
# Q-Q Plot: Energy Provided - Weekend
qqPlot(weekend_data$`Energy Provided (kWh)`, main = "Q-Q Plot: Energy Provided (Weekend)
    ylab = "Energy Provided (kWh)")
```



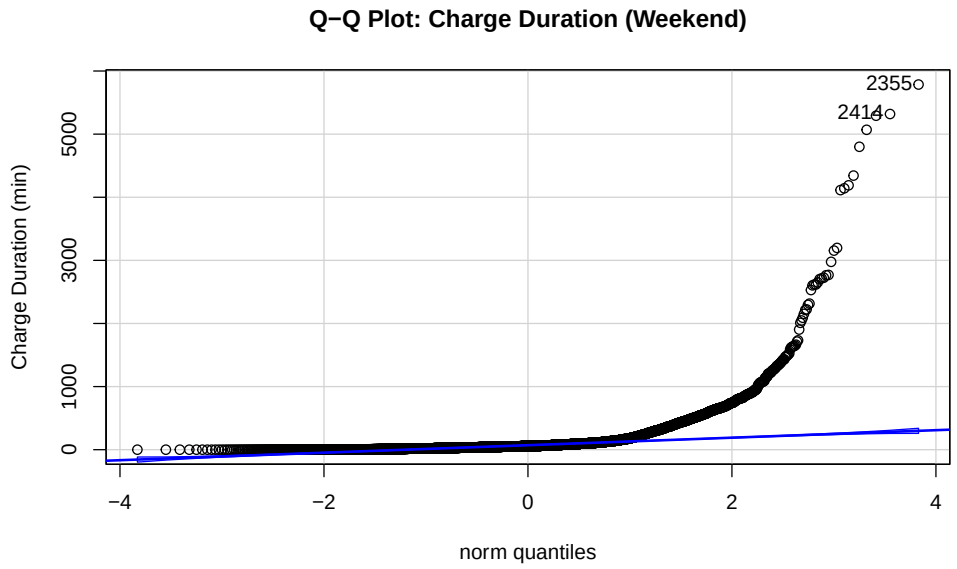

```
## [1] 4098 6547
```

```
# Q-Q Plot: Charge Duration - Weekday
qqPlot(weekday_data$`Charge Duration (min)`, main = "Q-Q Plot: Charge Duration (Weekday)
    ylab = "Charge Duration (min)")
```




```
## [1]  7027 11901
```

```r
# Q-Q Plot: Charge Duration - Weekend
qqPlot(weekend_data$`Charge Duration (min)`, main = "Q-Q Plot: Charge Duration (Weekend)",
    ylab = "Charge Duration (min)")
```




```
## [1] 2355 2414
```

```r
#----------------------------------------
# 3. Formal Normality Test (Shapiro-Wilk)
#----------------------------------------
# Note: Shapiro-Wilk limited to 5000 observations, so we
# sample

set.seed(123)
shapiro_weekday_energy <- shapiro.test(sample(weekday_data$`Energy Provided (kWh)`,
    5000))
shapiro_weekend_energy <- shapiro.test(sample(weekend_data$`Energy Provided (kWh)`,
    min(5000, nrow(weekend_data))))
shapiro_weekday_duration <- shapiro.test(sample(weekday_data$`Charge Duration (min)`,
    5000))
shapiro_weekend_duration <- shapiro.test(sample(weekend_data$`Charge Duration (min)`,
    min(5000, nrow(weekend_data))))

cat("\n--- Shapiro-Wilk Normality Tests (  =", alpha, ") ---\n")
```

```
##
## --- Shapiro-Wilk Normality Tests (  = 0.05 ) ---
```

```r
cat("H0: Data is normally distributed\n")
```

```
## H0: Data is normally distributed
```

```r
cat("Ha: Data is not normally distributed\n\n")
```

```
## Ha: Data is not normally distributed
```

```r
cat("Energy (Weekday): W =", round(shapiro_weekday_energy$statistic,
    4), ", p-value =", format(shapiro_weekday_energy$p.value,
    scientific = TRUE), "→", ifelse(shapiro_weekday_energy$p.value <
    alpha, "Reject H0 (Non-normal)", "Fail to reject H0"), "\n")
```

```
## Energy (Weekday): W = 0.9451 , p-value = 1.155929e-39 → Reject H0 (Non-normal)
```

```r
cat("Energy (Weekend): W =", round(shapiro_weekend_energy$statistic,
    4), ", p-value =", format(shapiro_weekend_energy$p.value,
    scientific = TRUE), "→", ifelse(shapiro_weekend_energy$p.value <
    alpha, "Reject H0 (Non-normal)", "Fail to reject H0"), "\n")
```

```
## Energy (Weekend): W = 0.8901 , p-value = 8.185068e-51 → Reject H0 (Non-normal)
```

```r
cat("Duration (Weekday): W =", round(shapiro_weekday_duration$statistic,
    4), ", p-value =", format(shapiro_weekday_duration$p.value,
    scientific = TRUE), "→", ifelse(shapiro_weekday_duration$p.value <
    alpha, "Reject H0 (Non-normal)", "Fail to reject H0"), "\n")
```

```
## Duration (Weekday): W = 0.4702 , p-value = 2.087111e-81 → Reject H0 (Non-normal)
```

```r
cat("Duration (Weekend): W =", round(shapiro_weekend_duration$statistic,
    4), ", p-value =", format(shapiro_weekend_duration$p.value,
    scientific = TRUE), "→", ifelse(shapiro_weekend_duration$p.value <
    alpha, "Reject H0 (Non-normal)", "Fail to reject H0"), "\n")
```

```
## Duration (Weekend): W = 0.383 , p-value = 9.244873e-85 → Reject H0 (Non-normal)
```

```r
#----------------------------------------
# 4. Hypothesis Tests for Energy Provided
#----------------------------------------

cat("\n===== ENERGY PROVIDED: Weekend vs Weekday =====\n")
```

```
##
## ===== ENERGY PROVIDED: Weekend vs Weekday =====
```

```
cat("H0:  _weekday =  _weekend (No difference in mean energy)\n")
```

```
## H0:  _weekday =  _weekend (No difference in mean energy)
```

```
cat("Ha:  _weekday   _weekend (Difference exists)\n")
```

```
## Ha:  _weekday   _weekend (Difference exists)
```

```
cat("  =", alpha, "\n")
```

```
##   = 0.05
```

```
# Descriptive comparison
cat("\nDescriptive Statistics:\n")
```

```
##
## Descriptive Statistics:
```

```
cat("Weekday - Mean:", round(mean(weekday_data$`Energy Provided (kWh)`),
    3), "kWh, Median:", round(median(weekday_data$`Energy Provided (kWh)`),
    3), "kWh\n")
```

```
## Weekday - Mean: 27.151 kWh, Median: 24.106 kWh
```

```
cat("Weekend - Mean:", round(mean(weekend_data$`Energy Provided (kWh)`),
    3), "kWh, Median:", round(median(weekend_data$`Energy Provided (kWh)`),
    3), "kWh\n")
```

```
## Weekend - Mean: 27.562 kWh, Median: 25.14 kWh
```

```
# Two-sample t-test (Welch's t-test)
t_test_energy <- t.test(`Energy Provided (kWh)` ~ is_weekend,
    data = ev_data, conf.level = 1 - alpha)
cat("\n--- Two-Sample t-Test (Welch) ---\n")
```

```
##
## --- Two-Sample t-Test (Welch) ---
```

```r
cat("t-statistic:", round(t_test_energy$statistic, 4), "\n")
```

```
## t-statistic: -1.4153
```

```r
cat("Degrees of freedom:", round(t_test_energy$parameter, 2),
    "\n")
```

```
## Degrees of freedom: 15099.4
```

```r
cat("p-value:", format(t_test_energy$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 1.570139e-01
```

```r
cat("95% CI for difference (Weekday - Weekend):", round(t_test_energy$conf.int[1],
    3), "to", round(t_test_energy$conf.int[2], 3), "kWh\n")
```

```
## 95% CI for difference (Weekday - Weekend): -0.981 to 0.158 kWh
```

```r
cat("Decision:", ifelse(t_test_energy$p.value < alpha, "Reject H0 - Significant differen
    "Fail to reject H0 - No significant difference"), "\n")
```

```
## Decision: Fail to reject H0 - No significant difference
```

```r
# Wilcoxon rank-sum test (non-parametric alternative)
wilcox_energy <- wilcox.test(`Energy Provided (kWh)` ~ is_weekend,
    data = ev_data, conf.int = TRUE, conf.level = 1 - alpha)
cat("\n--- Wilcoxon Rank-Sum Test (Non-parametric) ---\n")
```

```
##
## --- Wilcoxon Rank-Sum Test (Non-parametric) ---
```

```r
cat("W statistic:", wilcox_energy$statistic, "\n")
```

```
## W statistic: 60137316
```

```r
cat("p-value:", format(wilcox_energy$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 2.021458e-01
```

```r
cat("95% CI for difference:", round(wilcox_energy$conf.int[1],
    3), "to", round(wilcox_energy$conf.int[2], 3), "kWh\n")
```

```
## 95% CI for difference: -0.78 to 0.149 kWh
```

```r
cat("Decision:", ifelse(wilcox_energy$p.value < alpha, "Reject H0 - Significant differen
    "Fail to reject H0 - No significant difference"), "\n")
```

```
## Decision: Fail to reject H0 - No significant difference
```

```r
#----------------------------------------
# 5. Hypothesis Tests for Charge Duration
#----------------------------------------

cat("\n===== CHARGE DURATION: Weekend vs Weekday =====\n")
```

```
##
## ===== CHARGE DURATION: Weekend vs Weekday =====
```

```r
cat("H0:  _weekday =  _weekend (No difference in mean duration)\n")
```

```
## H0:  _weekday =  _weekend (No difference in mean duration)
```

```r
cat("Ha:  _weekday   _weekend (Difference exists)\n")
```

```
## Ha:  _weekday   _weekend (Difference exists)
```

```r
cat("  =", alpha, "\n")
```

```
##   = 0.05
```

```r
# Descriptive comparison
cat("\nDescriptive Statistics:\n")
```

```
##
## Descriptive Statistics:
```

```r
cat("Weekday - Mean:", round(mean(weekday_data$`Charge Duration (min)`),
    3), "min, Median:", round(median(weekday_data$`Charge Duration (min)`),
    3), "min\n")
```

```
## Weekday - Mean: 130.035 min, Median: 58.917 min
```

```r
cat("Weekend - Mean:", round(mean(weekend_data$`Charge Duration (min)`),
    3), "min, Median:", round(median(weekend_data$`Charge Duration (min)`),
    3), "min\n")
```

```
## Weekend - Mean: 130.786 min, Median: 56.747 min
```

```r
# Two-sample t-test
t_test_duration <- t.test(`Charge Duration (min)` ~ is_weekend,
    data = ev_data, conf.level = 1 - alpha)
cat("\n--- Two-Sample t-Test (Welch) ---\n")
```

```
##
## --- Two-Sample t-Test (Welch) ---
```

```r
cat("t-statistic:", round(t_test_duration$statistic, 4), "\n")
```

```
## t-statistic: -0.2013
```

```r
cat("Degrees of freedom:", round(t_test_duration$parameter, 2),
    "\n")
```

```
## Degrees of freedom: 14563.67
```

```r
cat("p-value:", format(t_test_duration$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 8.405032e-01
```

```r
cat("95% CI for difference (Weekday - Weekend):", round(t_test_duration$conf.int[1],
    3), "to", round(t_test_duration$conf.int[2], 3), "min\n")
```

```
## 95% CI for difference (Weekday - Weekend): -8.06 to 6.559 min
```

```r
cat("Decision:", ifelse(t_test_duration$p.value < alpha, "Reject H0 - Significant differ
    "Fail to reject H0 - No significant difference"), "\n")
```

```
## Decision: Fail to reject H0 - No significant difference
```

```r
# Wilcoxon rank-sum test
wilcox_duration <- wilcox.test(`Charge Duration (min)` ~ is_weekend,
    data = ev_data, conf.int = TRUE, conf.level = 1 - alpha)
cat("\n--- Wilcoxon Rank-Sum Test (Non-parametric) ---\n")
```

```
##
## --- Wilcoxon Rank-Sum Test (Non-parametric) ---
```

```r
cat("W statistic:", wilcox_duration$statistic, "\n")
```

```
## W statistic: 61787124
```

```r
cat("p-value:", format(wilcox_duration$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 3.448716e-02
```

```r
cat("95% CI for difference:", round(wilcox_duration$conf.int[1],
    3), "to", round(wilcox_duration$conf.int[2], 3), "min\n")
```

```
## 95% CI for difference: 0.1 to 2.85 min
```

```r
cat("Decision:", ifelse(wilcox_duration$p.value < alpha, "Reject H0 - Significant differ
    "Fail to reject H0 - No significant difference"), "\n")
```

```
## Decision: Reject H0 - Significant difference exists
```

```
#----------------------------------------
# 6. Summary Table
#----------------------------------------
results_summary <- data.frame(Variable = c("Energy Provided (kWh)",
    "Charge Duration (min)"), Weekday_Mean = c(round(mean(weekday_data$`Energy Provided
    2), round(mean(weekday_data$`Charge Duration (min)`), 2)),
    Weekend_Mean = c(round(mean(weekend_data$`Energy Provided (kWh)`),
        2), round(mean(weekend_data$`Charge Duration (min)`),
        2)), t_statistic = c(round(t_test_energy$statistic, 3),
        round(t_test_duration$statistic, 3)), p_value = c(format(t_test_energy$p.value,
        scientific = TRUE), format(t_test_duration$p.value, scientific = TRUE)),
    CI_lower = c(round(t_test_energy$conf.int[1], 3), round(t_test_duration$conf.int[1],
        3)), CI_upper = c(round(t_test_energy$conf.int[2], 3),
        round(t_test_duration$conf.int[2], 3)), Decision = c(ifelse(t_test_energy$p.valu
        alpha, "Reject H0", "Fail to reject H0"), ifelse(t_test_duration$p.value <
        alpha, "Reject H0", "Fail to reject H0")))

kable(results_summary, caption = paste("Weekend vs Weekday Comparison: Two-Sample t-Test
    alpha, ")"), col.names = c("Variable", "Weekday Mean", "Weekend Mean",
    "t-statistic", "p-value", "95% CI Lower", "95% CI Upper",
    "Decision"), align = "lrrrrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 3: Weekend vs Weekday Comparison: Two-Sample t-Test Results (= 0.05)

Variable	Weekday Mean	Weekend Mean	t-statistic	p-value	95% CI Lower
Energy Provided (kWh)	27.15	27.56	-1.415	1.570139e-01	-0.981
Charge Duration (min)	130.04	130.79	-0.201	8.405032e-01	-8.060

```
### Location Differences Analysis Significance Level:   =
### 0.05

alpha <- 0.05

#----------------------------------------
# 1. Select Top 4 Locations by Frequency
#----------------------------------------

top_locations <- ev_data %>%
    count(`Location Name`, sort = TRUE) %>%
    head(4) %>%
```

```
    pull(`Location Name`)

cat("Top 4 Locations by Number of Sessions:\n")
```

```
## Top 4 Locations by Number of Sessions:
```

```
ev_data %>%
    count(`Location Name`, sort = TRUE) %>%
    head(4) %>%
    print()
```

```
##                                 Location Name    n
## 1  Delancey and Essex Municipal Parking Garage 9030
## 2        Court Square Municipal Parking Garage 8357
## 3 Queens Borough Hall Municipal Parking Garage 3228
## 4 Jerome 190th Street Municipal Parking Garage  832
```

```
# Filter data to top 4 locations
ev_top_locations <- ev_data %>%
    filter(`Location Name` %in% top_locations)

cat("\nFiltered dataset size:", nrow(ev_top_locations), "sessions\n")
```

```
##
## Filtered dataset size: 21447 sessions
```

```
#----------------------------------------
# 2. Descriptive Statistics by Location
#----------------------------------------

location_summary <- ev_top_locations %>%
    group_by(`Location Name`) %>%
    summarise(n = n(), Mean = round(mean(`Energy Provided (kWh)`,
        na.rm = TRUE), 2), Median = round(median(`Energy Provided (kWh)`,
        na.rm = TRUE), 2), SD = round(sd(`Energy Provided (kWh)`,
        na.rm = TRUE), 2), Variance = round(var(`Energy Provided (kWh)`,
        na.rm = TRUE), 2), .groups = "drop")

kable(location_summary, caption = "Energy Provided (kWh) by Location - Descriptive Stati
    col.names = c("Location", "n", "Mean", "Median", "Std Dev",
        "Variance"), align = "lrrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 4: Energy Provided (kWh) by Location - Descriptive Statistics

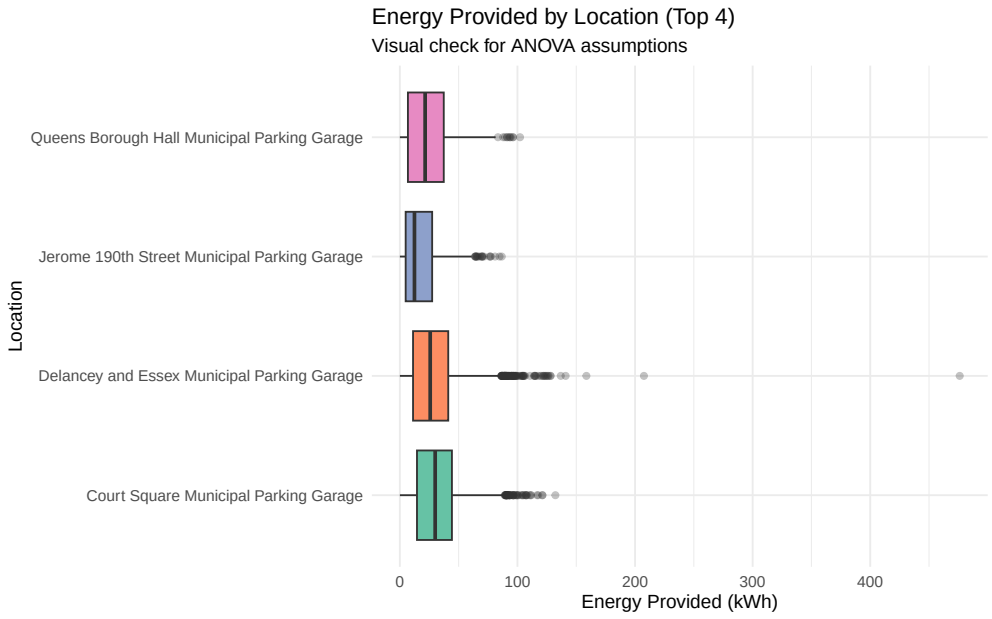
Location	n	Mean	Median	Std Dev	Variance
Court Square Municipal Parking Garage	8357	31.41	30.04	19.97	398.63
Delancey and Essex Municipal Parking Garage	9030	28.44	25.80	21.54	464.13
Jerome 190th Street Municipal Parking Garage	832	18.22	12.42	17.26	297.76
Queens Borough Hall Municipal Parking Garage	3228	24.07	21.50	19.40	376.19

```
#----------------------------------------
# 3. ANOVA Assumption Checks
#----------------------------------------

cat("\n===== ANOVA ASSUMPTION CHECKS =====\n")
```

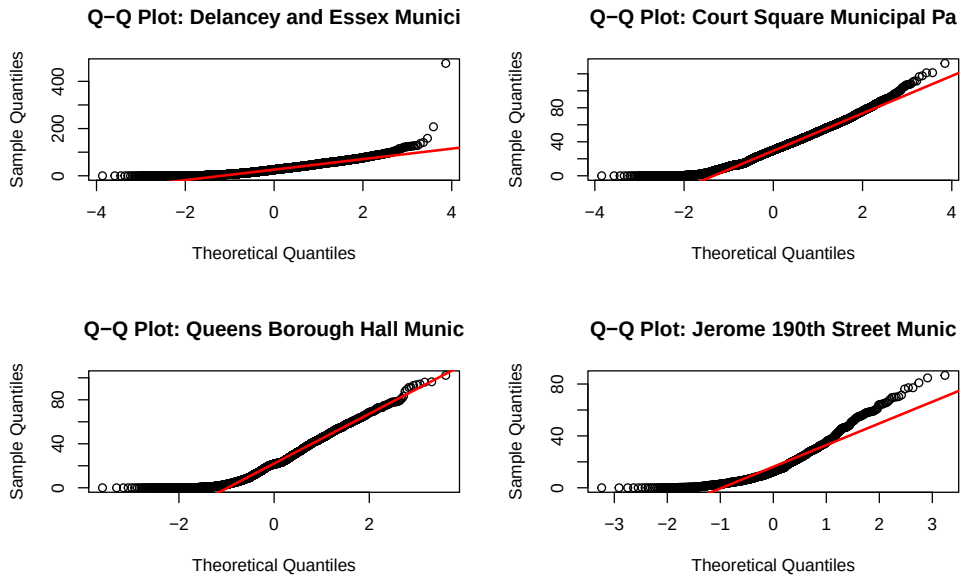
```
##
## ===== ANOVA ASSUMPTION CHECKS =====
```

```
# 3a. Boxplots - Visual check for outliers and distribution
ggplot(ev_top_locations, aes(x = `Location Name`, y = `Energy Provided (kWh)`,
    fill = `Location Name`)) + geom_boxplot(outlier.alpha = 0.3) +
    coord_flip() + labs(title = "Energy Provided by Location (Top 4)",
    subtitle = "Visual check for ANOVA assumptions", x = "Location",
    y = "Energy Provided (kWh)") + theme_minimal() + theme(legend.position = "none") +
    scale_fill_brewer(palette = "Set2")
```



```r
# 3b. Q-Q Plots for each location
par(mfrow = c(2, 2))
for (loc in top_locations) {
    loc_data <- ev_top_locations %>%
        filter(`Location Name` == loc) %>%
        pull(`Energy Provided (kWh)`)

    qqnorm(loc_data, main = paste("Q-Q Plot:", substr(loc, 1,
        25)))
    qqline(loc_data, col = "red", lwd = 2)
}
```




```r
par(mfrow = c(1, 1))

# 3c. Shapiro-Wilk Test for Normality (sample due to size
# limit)
cat("\n--- Shapiro-Wilk Normality Tests (  =", alpha, ") ---\n")
```

```
##
## --- Shapiro-Wilk Normality Tests (  = 0.05 ) ---
```

```r
cat("H0: Data is normally distributed\n")
```

```
## H0: Data is normally distributed
```

```r
cat("Ha: Data is not normally distributed\n\n")
```

```
## Ha: Data is not normally distributed
```

```r
set.seed(123)
for (loc in top_locations) {
    loc_data <- ev_top_locations %>%
        filter(`Location Name` == loc) %>%
        pull(`Energy Provided (kWh)`)

    sample_size <- min(5000, length(loc_data))
    shapiro_result <- shapiro.test(sample(loc_data, sample_size))

    cat(substr(loc, 1, 40), ":\n")
    cat("  W =", round(shapiro_result$statistic, 4), ", p-value =",
        format(shapiro_result$p.value, scientific = TRUE), "→",
        ifelse(shapiro_result$p.value < alpha, "Non-normal",
            "Normal"), "\n")
}
```

```
## Delancey and Essex Municipal Parking Gar :
##   W = 0.8893 , p-value = 6.101835e-51 → Non-normal
## Court Square Municipal Parking Garage :
##   W = 0.9718 , p-value = 2.818884e-30 → Non-normal
## Queens Borough Hall Municipal Parking Ga :
##   W = 0.9345 , p-value = 1.599682e-35 → Non-normal
## Jerome 190th Street Municipal Parking Ga :
##   W = 0.8691 , p-value = 8.623545e-26 → Non-normal
```

```r
# 3d. Levene's Test for Homogeneity of Variances
cat("\n--- Levene's Test for Homogeneity of Variances ---\n")
```

```
##
## --- Levene's Test for Homogeneity of Variances ---
```

```r
cat("H0: Variances are equal across groups\n")
```

```
## H0: Variances are equal across groups
```

```r
cat("Ha: Variances are not equal\n\n")
```

```
## Ha: Variances are not equal
```

```r
levene_result <- leveneTest(`Energy Provided (kWh)` ~ `Location Name`,
    data = ev_top_locations)
cat("F-statistic:", round(levene_result$`F value`[1], 4), "\n")
```

```
## F-statistic: 26.9884
```

```r
cat("p-value:", format(levene_result$`Pr(>F)`[1], scientific = TRUE),
    "\n")
```

```
## p-value: 2.052493e-17
```

```r
cat("Decision:", ifelse(levene_result$`Pr(>F)`[1] < alpha, "Reject H0 - Variances are NO
    "Fail to reject H0 - Variances are equal"), "\n")
```

```
## Decision: Reject H0 - Variances are NOT equal
```

```r
#----------------------------------------
# 4. One-Way ANOVA
#----------------------------------------

cat("\n===== ONE-WAY ANOVA =====\n")
```

```
##
## ===== ONE-WAY ANOVA =====
```

```r
cat("H0:  1 =  2 =  3 =  4 (All location means are equal)\n")
```

```
## H0:  1 =  2 =  3 =  4 (All location means are equal)
```

```r
cat("Ha: At least one mean is different\n")
```

```
## Ha: At least one mean is different
```

```r
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Fix: Create a clean column name for ANOVA
ev_top_locations <- ev_top_locations %>%
    mutate(Location = `Location Name`)

# Run ANOVA with clean column name
anova_model <- aov(`Energy Provided (kWh)` ~ Location, data = ev_top_locations)
anova_result <- summary(anova_model)

print(anova_result)
```

```
##                Df  Sum Sq Mean Sq F value Pr(>F)
## Location        3  221992   73997   176.6 <2e-16 ***
## Residuals   21443 8982976     419
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Extract F-statistic and p-value
f_stat <- anova_result[[1]]$`F value`[1]
p_value_anova <- anova_result[[1]]$`Pr(>F)`[1]

cat("\nF-statistic:", round(f_stat, 4), "\n")
```

```
##
## F-statistic: 176.6367
```

```r
cat("p-value:", format(p_value_anova, scientific = TRUE), "\n")
```

```
## p-value: 3.889395e-113
```

```r
cat("Decision:", ifelse(p_value_anova < alpha, "Reject H0 - Significant difference exist
    "Fail to reject H0 - No significant difference"), "\n")
```

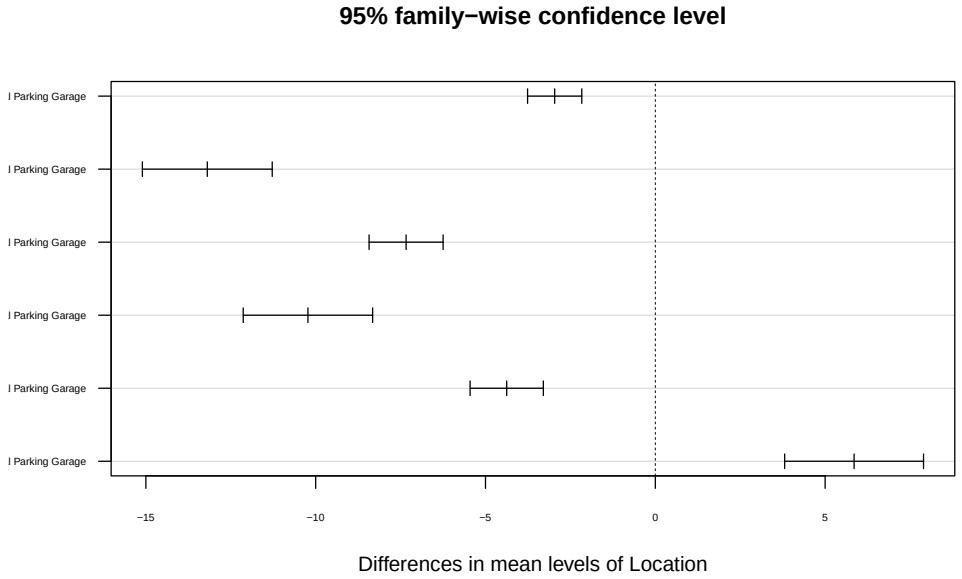
```
## Decision: Reject H0 - Significant difference exists between locations
```

```r
# Post-hoc Tukey HSD (now works with clean column name)
if (p_value_anova < alpha) {
    cat("\n--- Tukey HSD Post-Hoc Test ---\n")
    tukey_result <- TukeyHSD(anova_model, conf.level = 1 - alpha)
    print(tukey_result)

    # Visualize Tukey results
    plot(tukey_result, las = 1, cex.axis = 0.5)
}
```

```
##
## --- Tukey HSD Post-Hoc Test ---
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = `Energy Provided (kWh)` ~ Location, data = ev_top_locations)
##
## $Location
##
## Delancey and Essex Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Court Square Municipal Parking Garage
## Queens Borough Hall Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Jerome 190th Street Municipal Parking Ga
##
## Delancey and Essex Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Court Square Municipal Parking Garage
## Queens Borough Hall Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Jerome 190th Street Municipal Parking Ga
##
## Delancey and Essex Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Court Square Municipal Parking Garage
## Queens Borough Hall Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Jerome 190th Street Municipal Parking Ga
##
## Delancey and Essex Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Court Square Municipal Parking Garage
## Queens Borough Hall Municipal Parking Garage-Court Square Municipal Parking Garage
## Jerome 190th Street Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
```

```
## Queens Borough Hall Municipal Parking Garage-Delancey and Essex Municipal Parking Gar
## Queens Borough Hall Municipal Parking Garage-Jerome 190th Street Municipal Parking Ga
```




```
#----------------------------------------
# 5. Kruskal-Wallis Test (Non-parametric alternative)
#----------------------------------------

cat("\n===== KRUSKAL-WALLIS TEST (Non-parametric) =====\n")
```

```
##
## ===== KRUSKAL-WALLIS TEST (Non-parametric) =====
```

```
cat("H0: All location distributions are equal\n")
```

```
## H0: All location distributions are equal
```

```
cat("Ha: At least one distribution is different\n")
```

```
## Ha: At least one distribution is different
```

```
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
kruskal_result <- kruskal.test(`Energy Provided (kWh)` ~ `Location Name`,
    data = ev_top_locations)

cat("Chi-squared statistic:", round(kruskal_result$statistic,
    4), "\n")
```

```
## Chi-squared statistic: 625.3381
```

```r
cat("Degrees of freedom:", kruskal_result$parameter, "\n")
```

```
## Degrees of freedom: 3
```

```r
cat("p-value:", format(kruskal_result$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 3.237799e-135
```

```r
cat("Decision:", ifelse(kruskal_result$p.value < alpha, "Reject H0 - Significant differe
    "Fail to reject H0 - No significant difference"), "\n")
```

```
## Decision: Reject H0 - Significant difference exists between locations
```

```r
# Post-hoc Dunn Test (if Kruskal-Wallis is significant)
if (kruskal_result$p.value < alpha) {
    cat("\n--- Dunn's Post-Hoc Test (Bonferroni adjustment) ---\n")
    # Need to install FSA package for dunnTest
    if (!require(FSA))
        install.packages("FSA", quiet = TRUE)
    library(FSA)
    dunn_result <- dunnTest(`Energy Provided (kWh)` ~ `Location Name`,
        data = ev_top_locations, method = "bonferroni")
    print(dunn_result)
}
```

```
##
## --- Dunn's Post-Hoc Test (Bonferroni adjustment) ---
```

```
##                                                                                 Co
## 1         Court Square Municipal Parking Garage - Delancey and Essex Municipal Parkin
## 2        Court Square Municipal Parking Garage - Jerome 190th Street Municipal Parkin
## 3  Delancey and Essex Municipal Parking Garage - Jerome 190th Street Municipal Parkin
```

```
## 4        Court Square Municipal Parking Garage - Queens Borough Hall Municipal Parkin
## 5  Delancey and Essex Municipal Parking Garage - Queens Borough Hall Municipal Parkin
## 6 Jerome 190th Street Municipal Parking Garage - Queens Borough Hall Municipal Parkin
##           Z      P.unadj        P.adj
## 1 11.540782 8.217480e-31 4.930488e-30
## 2 19.297115 5.679322e-83 3.407593e-82
## 3 14.527499 8.112834e-48 4.867700e-47
## 4 18.642489 1.453203e-77 8.719217e-77
## 5 10.296738 7.289087e-25 4.373452e-24
## 6 -8.106531 5.208534e-16 3.125121e-15
```

```
#----------------------------------------
# 6. Chi-Square Test of Independence
#----------------------------------------

cat("\n===== CHI-SQUARE TEST OF INDEPENDENCE =====\n")
```

```
##
## ===== CHI-SQUARE TEST OF INDEPENDENCE =====
```

```
cat("H0: Location and Energy Category are independent\n")
```

```
## H0: Location and Energy Category are independent
```

```
cat("Ha: Location and Energy Category are dependent\n")
```

```
## Ha: Location and Energy Category are dependent
```

```
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```
# Create energy categories for contingency table
ev_top_locations <- ev_top_locations %>%
    mutate(Energy_Category = cut(`Energy Provided (kWh)`, breaks = c(-Inf,
        10, 25, 50, Inf), labels = c("Low (0-10)", "Medium (10-25)",
        "High (25-50)", "Very High (50+)")))

# Create contingency table
contingency_table <- table(ev_top_locations$`Location Name`,
    ev_top_locations$Energy_Category)

cat("Contingency Table:\n")
```

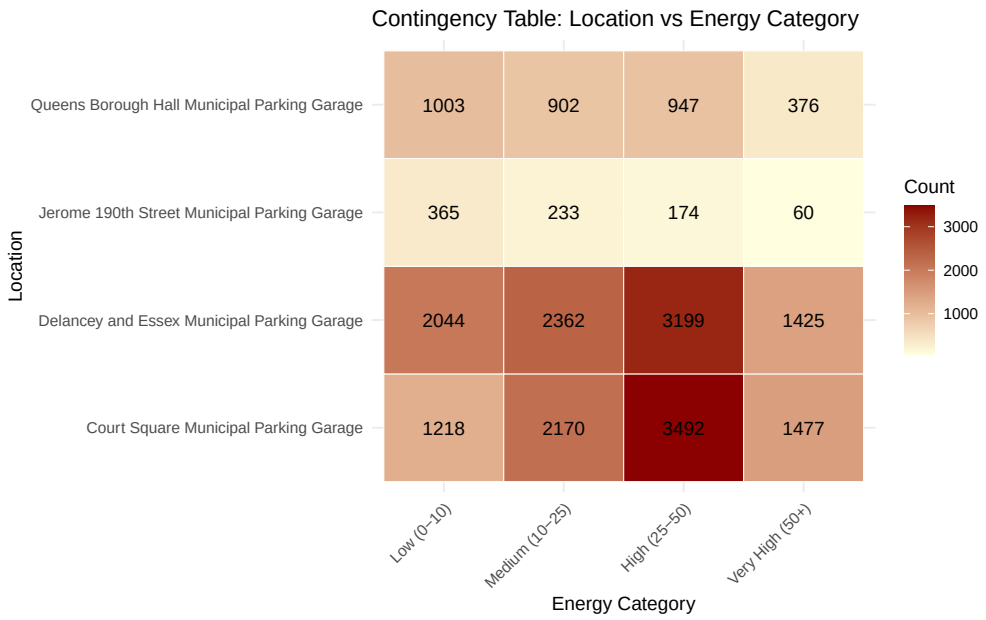
Contingency Table:

```
print(contingency_table)
```

```
##
##                                                 Low (0-10) Medium (10-25)
##   Court Square Municipal Parking Garage               1218            2170
##   Delancey and Essex Municipal Parking Garage         2044            2362
##   Jerome 190th Street Municipal Parking Garage         365             233
##   Queens Borough Hall Municipal Parking Garage        1003             902
##
##                                                 High (25-50) Very High (50+)
##   Court Square Municipal Parking Garage                 3492            1477
##   Delancey and Essex Municipal Parking Garage           3199            1425
##   Jerome 190th Street Municipal Parking Garage           174              60
##   Queens Borough Hall Municipal Parking Garage           947             376
```

```
# Visualize contingency table
contingency_df <- as.data.frame(contingency_table)
names(contingency_df) <- c("Location", "Energy_Category", "Count")

ggplot(contingency_df, aes(x = Energy_Category, y = Location,
    fill = Count)) + geom_tile(color = "white") + geom_text(aes(label = Count),
    size = 4) + scale_fill_gradient(low = "lightyellow", high = "darkred") +
    labs(title = "Contingency Table: Location vs Energy Category",
        x = "Energy Category", y = "Location") + theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```r
# Chi-square test
chi_sq_result <- chisq.test(contingency_table)

cat("\n--- Chi-Square Test Results ---\n")
```

```
##
## --- Chi-Square Test Results ---
```

```r
cat("Chi-squared statistic:", round(chi_sq_result$statistic,
    4), "\n")
```

```
## Chi-squared statistic: 786.4998
```

```r
cat("Degrees of freedom:", chi_sq_result$parameter, "\n")
```

```
## Degrees of freedom: 9
```

```r
cat("p-value:", format(chi_sq_result$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 1.71125e-163
```

```r
cat("Decision:", ifelse(chi_sq_result$p.value < alpha, "Reject H0 - Location and Energy
    "Fail to reject H0 - Location and Energy Category are independent"),
    "\n")
```

```
## Decision: Reject H0 - Location and Energy Category are DEPENDENT
```

```r
# Expected frequencies (check assumption: all expected
# counts > 5)
cat("\nExpected Frequencies:\n")
```

```
##
## Expected Frequencies:
```

```r
print(round(chi_sq_result$expected, 2))
```

```
##
##                                                 Low (0-10) Medium (10-25)
##   Court Square Municipal Parking Garage            1804.12        2208.19
##   Delancey and Essex Municipal Parking Garage      1949.41        2386.02
##   Jerome 190th Street Municipal Parking Garage      179.61         219.84
##   Queens Borough Hall Municipal Parking Garage      696.86         852.94
##
##                                                 High (25-50) Very High (50+)
##   Court Square Municipal Parking Garage              3044.01         1300.68
##   Delancey and Essex Municipal Parking Garage        3289.15         1405.42
##   Jerome 190th Street Municipal Parking Garage        303.05          129.49
##   Queens Borough Hall Municipal Parking Garage       1175.79          502.40
```

```
cat("\nAssumption Check: All expected frequencies > 5?\n")
```

```
##
## Assumption Check: All expected frequencies > 5?
```

```
cat(ifelse(all(chi_sq_result$expected >= 5), "Yes - Assumption met",
    "No - Consider Fisher's exact test or combine categories"),
    "\n")
```

```
## Yes - Assumption met
```

```
#----------------------------------------
# 7. Summary Table
#----------------------------------------

summary_tests <- data.frame(Test = c("One-Way ANOVA", "Kruskal-Wallis",
    "Chi-Square Independence"), Statistic = c(paste("F =", round(f_stat,
    3)), paste(" ² =", round(kruskal_result$statistic, 3)),
    paste(" ² =", round(chi_sq_result$statistic, 3))), df = c(paste(anova_result[[1]]$Df
    "and", anova_result[[1]]$Df[2]), kruskal_result$parameter,
    chi_sq_result$parameter), p_value = c(format(p_value_anova,
    scientific = TRUE), format(kruskal_result$p.value, scientific = TRUE),
    format(chi_sq_result$p.value, scientific = TRUE)), Decision = c(ifelse(p_value_anova
    alpha, "Reject H0", "Fail to reject H0"), ifelse(kruskal_result$p.value <
    alpha, "Reject H0", "Fail to reject H0"), ifelse(chi_sq_result$p.value <
    alpha, "Reject H0", "Fail to reject H0")))

kable(summary_tests, caption = paste("Location Differences Analysis Summary (  =",
    alpha, ")"), col.names = c("Test", "Test Statistic", "df",
    "p-value", "Decision"), align = "llrrr") %>%
```

```
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 5: Location Differences Analysis Summary (= 0.05)

Test	Test Statistic	df	p-value	Decision
One-Way ANOVA	F = 176.637	3 and 21443	3.889395e-113	Reject H0
Kruskal-Wallis	^{2} = 625.338	3	3.237799e-135	Reject H0
Chi-Square Independence	^{2} = 786.5	9	1.71125e-163	Reject H0

Proportion inference

One-sample proportion CI and test for weekend sessions:

H0: weekend = 0.3 or some reference value.

Two-sample proportion test: compare weekend usage between two specific locations.

```
### Proportion Inference Significance Level:   = 0.05

alpha <- 0.05

#----------------------------------------
# 1. One-Sample Proportion: Weekend Sessions
#----------------------------------------

cat("===== ONE-SAMPLE PROPORTION TEST: Weekend Sessions =====\n")
```

```
## ===== ONE-SAMPLE PROPORTION TEST: Weekend Sessions =====
```

```
cat("H0: p_weekend = 0.30 (30% of sessions are on weekends)\n")
```

```
## H0: p_weekend = 0.30 (30% of sessions are on weekends)
```

```
cat("Ha: p_weekend   0.30\n")
```

```
## Ha: p_weekend   0.30
```

```
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Calculate observed proportion
n_total <- nrow(ev_data)
n_weekend <- sum(ev_data$is_weekend == TRUE)
p_observed <- n_weekend/n_total

cat("--- Sample Statistics ---\n")
```

```
## --- Sample Statistics ---
```

```r
cat("Total sessions:", n_total, "\n")
```

```
## Total sessions: 23391
```

```r
cat("Weekend sessions:", n_weekend, "\n")
```

```
## Weekend sessions: 7788
```

```r
cat("Observed proportion (p̂):", round(p_observed, 4), "\n")
```

```
## Observed proportion (p̂): 0.3329
```

```r
cat("Expected proportion (p0): 0.30\n\n")
```

```
## Expected proportion (p0): 0.30
```

```r
# One-sample proportion test
p0 <- 0.3  # Hypothesized proportion
prop_test_one <- prop.test(n_weekend, n_total, p = p0, conf.level = 1 -
    alpha, correct = FALSE)  # Without Yates continuity correction

cat("--- One-Sample z-Test for Proportion ---\n")
```

```
## --- One-Sample z-Test for Proportion ---
```

```r
cat("Chi-squared statistic:", round(prop_test_one$statistic,
    4), "\n")
```

```
## Chi-squared statistic: 120.9213
```

```r
cat("p-value:", format(prop_test_one$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 3.976055e-28
```

```r
cat("95% Confidence Interval for p_weekend:", round(prop_test_one$conf.int[1],
    4), "to", round(prop_test_one$conf.int[2], 4), "\n")
```

```
## 95% Confidence Interval for p_weekend: 0.3269 to 0.339
```

```r
cat("Decision:", ifelse(prop_test_one$p.value < alpha, "Reject H0 - Weekend proportion d
    "Fail to reject H0 - No evidence weekend proportion differs from 0.30"),
    "\n")
```

```
## Decision: Reject H0 - Weekend proportion differs from 0.30
```

```r
# Manual z-test calculation for clarity
z_stat <- (p_observed - p0)/sqrt(p0 * (1 - p0)/n_total)
p_value_z <- 2 * pnorm(-abs(z_stat))  # Two-tailed

cat("\n--- Manual z-Test Calculation ---\n")
```

```
##
## --- Manual z-Test Calculation ---
```

```r
cat("z-statistic:", round(z_stat, 4), "\n")
```

```
## z-statistic: 10.9964
```

```r
cat("p-value (two-tailed):", format(p_value_z, scientific = TRUE),
    "\n")
```

```
## p-value (two-tailed): 3.976055e-28
```

```r
# Wilson confidence interval (more accurate for
# proportions)
cat("\n--- Wilson 95% Confidence Interval ---\n")
```

```
##
## --- Wilson 95% Confidence Interval ---
```

```r
wilson_ci <- prop.test(n_weekend, n_total, conf.level = 1 - alpha,
    correct = FALSE)$conf.int
cat("95% CI:", round(wilson_ci[1], 4), "to", round(wilson_ci[2],
    4), "\n")
```

```
## 95% CI: 0.3269 to 0.339
```

```r
#----------------------------------------
# 2. Two-Sample Proportion: Weekend Usage by Location
#----------------------------------------

cat("\n\n===== TWO-SAMPLE PROPORTION TEST: Weekend Usage by Location =====\n")
```

```
##
##
## ===== TWO-SAMPLE PROPORTION TEST: Weekend Usage by Location =====
```

```r
# Select two locations for comparison (top 2 by frequency)
top_2_locations <- ev_data %>%
    count(`Location Name`, sort = TRUE) %>%
    head(2) %>%
    pull(`Location Name`)

cat("Comparing locations:\n")
```

```
## Comparing locations:
```

```r
cat("  Location 1:", top_2_locations[1], "\n")
```

```
##   Location 1: Delancey and Essex Municipal Parking Garage
```

```r
cat("  Location 2:", top_2_locations[2], "\n\n")
```

```
##   Location 2: Court Square Municipal Parking Garage
```

```r
# Filter data for these two locations
loc1_data <- ev_data %>%
    filter(`Location Name` == top_2_locations[1])
loc2_data <- ev_data %>%
    filter(`Location Name` == top_2_locations[2])
```

```r
# Calculate proportions
n1 <- nrow(loc1_data)
n2 <- nrow(loc2_data)
x1 <- sum(loc1_data$is_weekend == TRUE)
x2 <- sum(loc2_data$is_weekend == TRUE)
p1 <- x1/n1
p2 <- x2/n2

cat("--- Sample Statistics ---\n")
```

```
## --- Sample Statistics ---
```

```r
cat("Location 1 -", top_2_locations[1], ":\n")
```

```
## Location 1 - Delancey and Essex Municipal Parking Garage :
```

```r
cat("  Total sessions:", n1, "\n")
```

```
##   Total sessions: 9030
```

```r
cat("  Weekend sessions:", x1, "\n")
```

```
##   Weekend sessions: 3323
```

```r
cat("  Proportion weekend (p̂1):", round(p1, 4), "\n\n")
```

```
##   Proportion weekend (p̂1): 0.368
```

```r
cat("Location 2 -", top_2_locations[2], ":\n")
```

```
## Location 2 - Court Square Municipal Parking Garage :
```

```r
cat("  Total sessions:", n2, "\n")
```

```
##   Total sessions: 8357
```

```r
cat("  Weekend sessions:", x2, "\n")
```

```
##   Weekend sessions: 2740
```

```r
cat("  Proportion weekend (p̂2):", round(p2, 4), "\n\n")
```

```
##   Proportion weekend (p̂2): 0.3279
```

```r
cat("Difference (p̂1 - p̂2):", round(p1 - p2, 4), "\n\n")
```

```
## Difference (p̂1 - p̂2): 0.0401
```

```r
# Hypothesis test
cat("H0: p1 = p2 (Weekend proportion is equal at both locations)\n")
```

```
## H0: p1 = p2 (Weekend proportion is equal at both locations)
```

```r
cat("Ha: p1  p2 (Weekend proportions differ)\n")
```

```
## Ha: p1  p2 (Weekend proportions differ)
```

```r
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Two-sample proportion test
prop_test_two <- prop.test(c(x1, x2), c(n1, n2), conf.level = 1 -
    alpha, correct = FALSE)

cat("--- Two-Sample z-Test for Proportions ---\n")
```

```
## --- Two-Sample z-Test for Proportions ---
```

```r
cat("Chi-squared statistic:", round(prop_test_two$statistic,
    4), "\n")
```

```
## Chi-squared statistic: 30.7711
```

```r
cat("p-value:", format(prop_test_two$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 2.903336e-08
```

```r
cat("95% CI for difference (p1 - p2):", round(prop_test_two$conf.int[1],
    4), "to", round(prop_test_two$conf.int[2], 4), "\n")
```

```
## 95% CI for difference (p1 - p2): 0.026 to 0.0543
```

```r
cat("Decision:", ifelse(prop_test_two$p.value < alpha, "Reject H0 - Weekend proportions
    "Fail to reject H0 - No significant difference in weekend proportions"),
    "\n")
```

```
## Decision: Reject H0 - Weekend proportions differ between locations
```

```r
# Manual z-test calculation
p_pooled <- (x1 + x2)/(n1 + n2)
se_pooled <- sqrt(p_pooled * (1 - p_pooled) * (1/n1 + 1/n2))
z_stat_two <- (p1 - p2)/se_pooled
p_value_z_two <- 2 * pnorm(-abs(z_stat_two))

cat("\n--- Manual z-Test Calculation ---\n")
```

```
##
## --- Manual z-Test Calculation ---
```

```r
cat("Pooled proportion:", round(p_pooled, 4), "\n")
```

```
## Pooled proportion: 0.3487
```

```r
cat("Standard error:", round(se_pooled, 4), "\n")
```

```
## Standard error: 0.0072
```

```r
cat("z-statistic:", round(z_stat_two, 4), "\n")
```

```
## z-statistic: 5.5472
```

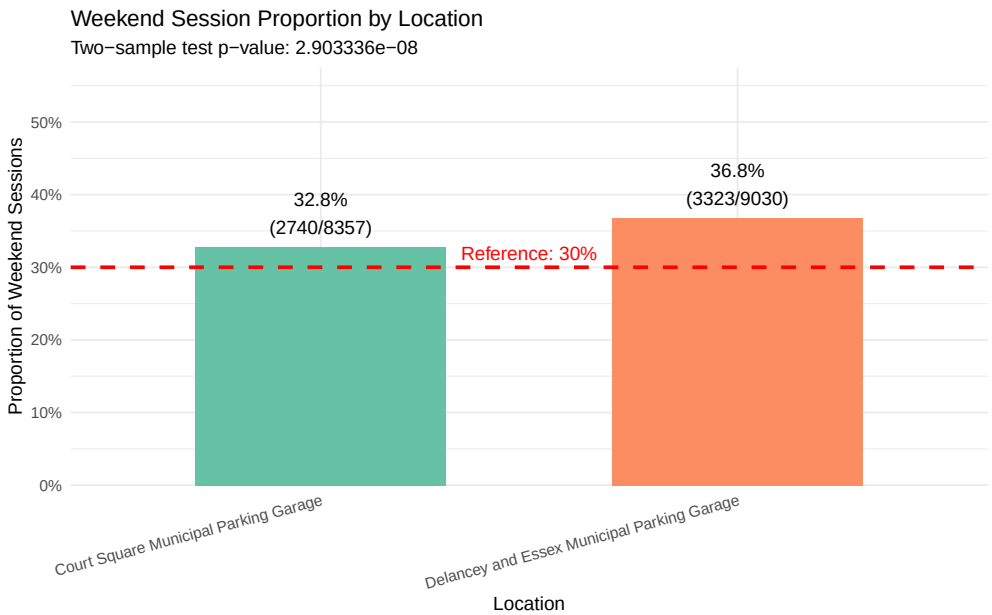
```r
cat("p-value (two-tailed):", format(p_value_z_two, scientific = TRUE),
    "\n")
```

```
## p-value (two-tailed): 2.903336e-08
```

```r
#----------------------------------------
# 3. Visualization
#----------------------------------------

# Bar plot comparing weekend proportions by location
prop_comparison <- data.frame(Location = c(top_2_locations[1],
    top_2_locations[2]), Weekend_Proportion = c(p1, p2), Weekend_Count = c(x1,
    x2), Total_Count = c(n1, n2))

ggplot(prop_comparison, aes(x = Location, y = Weekend_Proportion,
    fill = Location)) + geom_bar(stat = "identity", width = 0.6) +
    geom_text(aes(label = paste0(round(Weekend_Proportion * 100,
        1), "%\n(", Weekend_Count, "/", Total_Count, ")")), vjust = -0.3,
        size = 4) + geom_hline(yintercept = 0.3, linetype = "dashed",
    color = "red", linewidth = 1) + annotate("text", x = 1.5,
    y = 0.32, label = "Reference: 30%", color = "red") + scale_y_continuous(labels = sca
    limits = c(0, 0.5), expand = expansion(mult = c(0, 0.15))) +
    labs(title = "Weekend Session Proportion by Location", subtitle = paste("Two-sample
        format(prop_test_two$p.value, scientific = TRUE)), x = "Location",
        y = "Proportion of Weekend Sessions") + theme_minimal() +
    theme(legend.position = "none", axis.text.x = element_text(angle = 15,
        hjust = 1)) + scale_fill_brewer(palette = "Set2")
```



```r
#----------------------------------------
# 4. Summary Table
#----------------------------------------

summary_proportion <- data.frame(Test = c("One-Sample (Weekend = 0.30)",
    paste("Two-Sample (", substr(top_2_locations[1], 1, 15),
        " vs ", substr(top_2_locations[2], 1, 15), ")", sep = "")),
    Observed = c(paste0(round(p_observed * 100, 2), "%"), paste0(round(p1 *
        100, 2), "% vs ", round(p2 * 100, 2), "%")), Test_Statistic = c(paste(" ² =",
        round(prop_test_one$statistic, 3)), paste(" ² =", round(prop_test_two$statistic,
        3))), p_value = c(format(prop_test_one$p.value, scientific = TRUE),
        format(prop_test_two$p.value, scientific = TRUE)), CI_95 = c(paste0("(",
        round(prop_test_one$conf.int[1], 4), ", ", round(prop_test_one$conf.int[2],
            4), ")"), paste0("(", round(prop_test_two$conf.int[1],
        4), ", ", round(prop_test_two$conf.int[2], 4), ")")),
    Decision = c(ifelse(prop_test_one$p.value < alpha, "Reject H0",
        "Fail to reject H0"), ifelse(prop_test_two$p.value <
        alpha, "Reject H0", "Fail to reject H0")))

kable(summary_proportion, caption = paste("Proportion Inference Summary (  =",
    alpha, ")"), col.names = c("Test", "Observed Proportion",
    "Test Statistic", "p-value", "95% CI", "Decision"), align = "llrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 6: Proportion Inference Summary (= 0.05)

Test	Observed Proportion	Test Statistic	p-val
One-Sample (Weekend = 0.30)	33.29%	² = 120.921	3.976055e-
Two-Sample (Delancey and Es vs Court Square Mu)	36.8% vs 32.79%	² = 30.771	2.903336e-

Variance/spread comparisons

F-test for variance in Energy Provided (kWh) between two groups:

PAID vs ROAMING (Session Status) or weekend vs weekday.

```r
### Variance/Spread Comparisons (F-Test) Significance
### Level:   = 0.05

alpha <- 0.05

#----------------------------------------
# 1. F-Test: PAID vs ROAMING (Session Status)
```

```r
#----------------------------------------
```

```r
cat("===== F-TEST FOR VARIANCE: PAID vs ROAMING =====\n")
```

```
## ===== F-TEST FOR VARIANCE: PAID vs ROAMING =====
```

```r
cat("H0:  ²_PAID =  ²_ROAMING (Variances are equal)\n")
```

```
## H0:  ²_PAID =  ²_ROAMING (Variances are equal)
```

```r
cat("Ha:  ²_PAID   ²_ROAMING (Variances are not equal)\n")
```

```
## Ha:  ²_PAID   ²_ROAMING (Variances are not equal)
```

```r
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Filter data for PAID and ROAMING sessions
paid_data <- ev_data %>%
    filter(`Session Status` == "PAID")
roaming_data <- ev_data %>%
    filter(`Session Status` == "ROAMING")

# Sample statistics
n_paid <- nrow(paid_data)
n_roaming <- nrow(roaming_data)
var_paid <- var(paid_data$`Energy Provided (kWh)`, na.rm = TRUE)
var_roaming <- var(roaming_data$`Energy Provided (kWh)`, na.rm = TRUE)
sd_paid <- sd(paid_data$`Energy Provided (kWh)`, na.rm = TRUE)
sd_roaming <- sd(roaming_data$`Energy Provided (kWh)`, na.rm = TRUE)

cat("--- Sample Statistics ---\n")
```

```
## --- Sample Statistics ---
```

```r
cat("PAID Sessions:\n")
```

```
## PAID Sessions:
```

```r
cat("  n =", n_paid, "\n")
```

```
##   n = 22287
```

```r
cat("  Variance (s²) =", round(var_paid, 4), "kWh²\n")
```

```
##   Variance (s²) = 429.9238 kWh²
```

```r
cat("  Std Dev (s) =", round(sd_paid, 4), "kWh\n\n")
```

```
##   Std Dev (s) = 20.7346 kWh
```

```r
cat("ROAMING Sessions:\n")
```

```
## ROAMING Sessions:
```

```r
cat("  n =", n_roaming, "\n")
```

```
##   n = 1104
```

```r
cat("  Variance (s²) =", round(var_roaming, 4), "kWh²\n")
```

```
##   Variance (s²) = 393.6776 kWh²
```

```r
cat("  Std Dev (s) =", round(sd_roaming, 4), "kWh\n\n")
```

```
##   Std Dev (s) = 19.8413 kWh
```

```r
cat("Variance Ratio (s²_PAID / s²_ROAMING):", round(var_paid/var_roaming,
    4), "\n\n")
```

```
## Variance Ratio (s²_PAID / s²_ROAMING): 1.0921
```

```r
# F-test
f_test_status <- var.test(paid_data$`Energy Provided (kWh)`,
    roaming_data$`Energy Provided (kWh)`, conf.level = 1 - alpha)

cat("--- F-Test Results ---\n")
```

```
## --- F-Test Results ---
```

```r
cat("F-statistic:", round(f_test_status$statistic, 4), "\n")
```

```
## F-statistic: 1.0921
```

```r
cat("Degrees of freedom: df1 =", f_test_status$parameter[1],
    ", df2 =", f_test_status$parameter[2], "\n")
```

```
## Degrees of freedom: df1 = 22286 , df2 = 1103
```

```r
cat("p-value:", format(f_test_status$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 4.779705e-02
```

```r
cat("95% CI for variance ratio:", round(f_test_status$conf.int[1],
    4), "to", round(f_test_status$conf.int[2], 4), "\n")
```

```
## 95% CI for variance ratio: 1.0009 to 1.1876
```

```r
cat("Decision:", ifelse(f_test_status$p.value < alpha, "Reject H0 - Variances are signi
    "Fail to reject H0 - No significant difference in variances"),
    "\n")
```

```
## Decision: Reject H0 - Variances are significantly different
```

```r
#----------------------------------------
# 2. F-Test: Weekend vs Weekday
#----------------------------------------

cat("\n\n===== F-TEST FOR VARIANCE: Weekend vs Weekday =====\n")
```

```
##
##
## ===== F-TEST FOR VARIANCE: Weekend vs Weekday =====
```

```r
cat("H0:  ²_weekend =  ²_weekday (Variances are equal)\n")
```

```
## H0:  ²_weekend =  ²_weekday (Variances are equal)
```

```r
cat("Ha:  ²_weekend   ²_weekday (Variances are not equal)\n")
```

```
## Ha:  ²_weekend   ²_weekday (Variances are not equal)
```

```r
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Filter data
weekday_data <- ev_data %>%
    filter(is_weekend == FALSE)
weekend_data <- ev_data %>%
    filter(is_weekend == TRUE)

# Sample statistics
n_weekday <- nrow(weekday_data)
n_weekend <- nrow(weekend_data)
var_weekday <- var(weekday_data$`Energy Provided (kWh)`, na.rm = TRUE)
var_weekend <- var(weekend_data$`Energy Provided (kWh)`, na.rm = TRUE)
sd_weekday <- sd(weekday_data$`Energy Provided (kWh)`, na.rm = TRUE)
sd_weekend <- sd(weekend_data$`Energy Provided (kWh)`, na.rm = TRUE)

cat("--- Sample Statistics ---\n")
```

```
## --- Sample Statistics ---
```

```r
cat("Weekday Sessions:\n")
```

```
## Weekday Sessions:
```

```r
cat("  n =", n_weekday, "\n")
```

```
##   n = 15603
```

```r
cat("  Variance (s²) =", round(var_weekday, 4), "kWh²\n")
```

```
##   Variance (s²) = 418.7135 kWh²
```

```r
cat("  Std Dev (s) =", round(sd_weekday, 4), "kWh\n\n")
```

```
##   Std Dev (s) = 20.4625 kWh
```

```r
cat("Weekend Sessions:\n")
```

```
## Weekend Sessions:
```

```r
cat("  n =", n_weekend, "\n")
```

```
##   n = 7788
```

```r
cat("  Variance (s²) =", round(var_weekend, 4), "kWh²\n")
```

```
##   Variance (s²) = 448.4592 kWh²
```

```r
cat("  Std Dev (s) =", round(sd_weekend, 4), "kWh\n\n")
```

```
##   Std Dev (s) = 21.1769 kWh
```

```r
cat("Variance Ratio (s²_weekday / s²_weekend):", round(var_weekday/var_weekend,
    4), "\n\n")
```

```
## Variance Ratio (s²_weekday / s²_weekend): 0.9337
```

```r
# F-test
f_test_weekend <- var.test(weekday_data$`Energy Provided (kWh)`,
    weekend_data$`Energy Provided (kWh)`, conf.level = 1 - alpha)

cat("--- F-Test Results ---\n")
```

```
## --- F-Test Results ---
```

```r
cat("F-statistic:", round(f_test_weekend$statistic, 4), "\n")
```

```
## F-statistic: 0.9337
```

```r
cat("Degrees of freedom: df1 =", f_test_weekend$parameter[1],
    ", df2 =", f_test_weekend$parameter[2], "\n")
```

```
## Degrees of freedom: df1 = 15602 , df2 = 7787
```

```r
cat("p-value:", format(f_test_weekend$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 4.433402e-04
```

```r
cat("95% CI for variance ratio:", round(f_test_weekend$conf.int[1],
    4), "to", round(f_test_weekend$conf.int[2], 4), "\n")
```

```
## 95% CI for variance ratio: 0.8983 to 0.9702
```

```r
cat("Decision:", ifelse(f_test_weekend$p.value < alpha, "Reject H0 - Variances are signi
    "Fail to reject H0 - No significant difference in variances"),
    "\n")
```

```
## Decision: Reject H0 - Variances are significantly different
```

```r
#----------------------------------------
# 3. Levene's Test (Robust Alternative)
#----------------------------------------
# F-test assumes normality; Levene's test is more robust

cat("\n\n===== LEVENE'S TEST (Robust Alternative) =====\n")
```

```
##
##
## ===== LEVENE'S TEST (Robust Alternative) =====
```

```r
cat("More robust to departures from normality\n\n")
```

```
## More robust to departures from normality
```

```r
# Levene's Test: PAID vs ROAMING
cat("--- PAID vs ROAMING ---\n")
```

```
## --- PAID vs ROAMING ---
```

```r
ev_paid_roaming <- ev_data %>%
    filter(`Session Status` %in% c("PAID", "ROAMING"))
levene_status <- leveneTest(`Energy Provided (kWh)` ~ `Session Status`,
    data = ev_paid_roaming)
cat("F-statistic:", round(levene_status$`F value`[1], 4), "\n")
```

```
## F-statistic: 6.605
```

```r
cat("p-value:", format(levene_status$`Pr(>F)`[1], scientific = TRUE),
    "\n")
```

```
## p-value: 1.017548e-02
```

```r
cat("Decision:", ifelse(levene_status$`Pr(>F)`[1] < alpha, "Reject H0 - Variances differ
    "Fail to reject H0 - Variances equal"), "\n\n")
```

```
## Decision: Reject H0 - Variances differ
```

```r
# Levene's Test: Weekend vs Weekday
cat("--- Weekend vs Weekday ---\n")
```

```
## --- Weekend vs Weekday ---
```

```r
levene_weekend <- leveneTest(`Energy Provided (kWh)` ~ is_weekend,
    data = ev_data)
cat("F-statistic:", round(levene_weekend$`F value`[1], 4), "\n")
```

```
## F-statistic: 0.7101
```

```r
cat("p-value:", format(levene_weekend$`Pr(>F)`[1], scientific = TRUE),
    "\n")
```

```
## p-value: 3.994198e-01
```

```r
cat("Decision:", ifelse(levene_weekend$`Pr(>F)`[1] < alpha, "Reject H0 - Variances diffe
    "Fail to reject H0 - Variances equal"), "\n")
```

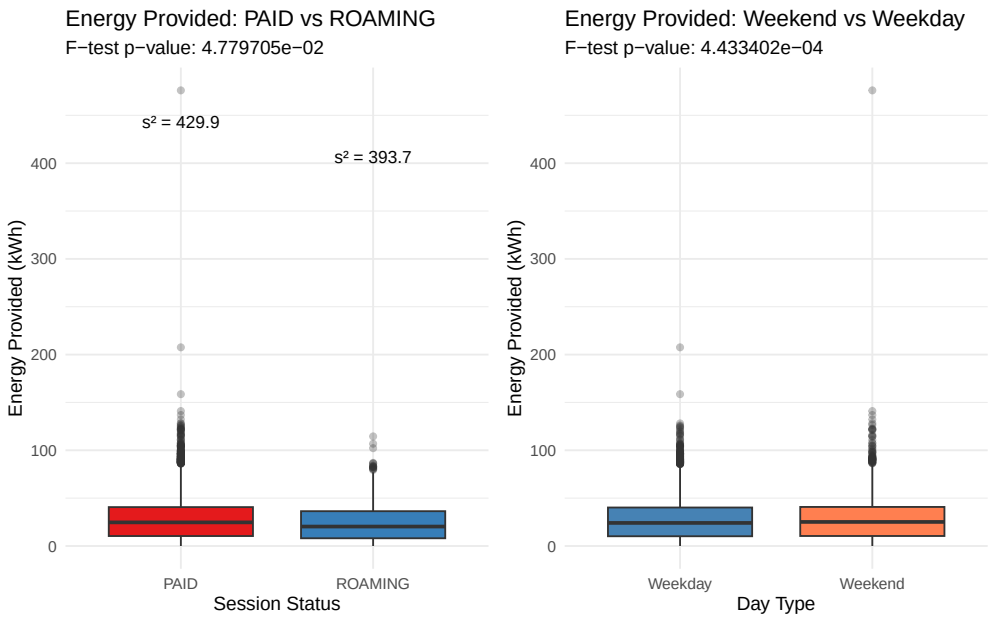
```
## Decision: Fail to reject H0 - Variances equal
```

```r
#----------------------------------------
# 4. Visualization
#----------------------------------------

# Boxplot: Energy by Session Status (PAID vs ROAMING)
p1 <- ggplot(ev_paid_roaming, aes(x = `Session Status`, y = `Energy Provided (kWh)`,
    fill = `Session Status`)) + geom_boxplot(outlier.alpha = 0.3) +
    stat_summary(fun = var, geom = "text", aes(label = paste("s² =",
        round(..y.., 1))), vjust = -0.5, size = 3.5) + labs(title = "Energy Provided: PA
    subtitle = paste("F-test p-value:", format(f_test_status$p.value,
        scientific = TRUE)), x = "Session Status", y = "Energy Provided (kWh)") +
    theme_minimal() + theme(legend.position = "none") + scale_fill_brewer(palette = "Se

# Boxplot: Energy by Weekend
p2 <- ggplot(ev_data, aes(x = ifelse(is_weekend, "Weekend", "Weekday"),
    y = `Energy Provided (kWh)`, fill = is_weekend)) + geom_boxplot(outlier.alpha = 0.3)
    labs(title = "Energy Provided: Weekend vs Weekday", subtitle = paste("F-test p-value
        format(f_test_weekend$p.value, scientific = TRUE)), x = "Day Type",
        y = "Energy Provided (kWh)") + theme_minimal() + theme(legend.position = "none")
    scale_fill_manual(values = c("steelblue", "coral"))

# Display plots side by side
grid.arrange(p1, p2, ncol = 2)
```




```r
#----------------------------------------
# 5. Summary Table
```

```
#----------------------------------------

summary_variance <- data.frame(Comparison = c("PAID vs ROAMING",
    "Weekend vs Weekday"), Group1_Variance = c(round(var_paid,
    2), round(var_weekday, 2)), Group2_Variance = c(round(var_roaming,
    2), round(var_weekend, 2)), F_statistic = c(round(f_test_status$statistic,
    4), round(f_test_weekend$statistic, 4)), df = c(paste(f_test_status$parameter[1],
    ",", f_test_status$parameter[2]), paste(f_test_weekend$parameter[1],
    ",", f_test_weekend$parameter[2])), p_value = c(format(f_test_status$p.value,
    scientific = TRUE), format(f_test_weekend$p.value, scientific = TRUE)),
    CI_95 = c(paste0("(", round(f_test_status$conf.int[1], 3),
        ", ", round(f_test_status$conf.int[2], 3), ")"), paste0("(",
        round(f_test_weekend$conf.int[1], 3), ", ", round(f_test_weekend$conf.int[2],
            3), ")")), Decision = c(ifelse(f_test_status$p.value <
        alpha, "Reject H0", "Fail to reject H0"), ifelse(f_test_weekend$p.value <
        alpha, "Reject H0", "Fail to reject H0")))

kable(summary_variance, caption = paste("F-Test for Variance Comparison (  =",
    alpha, ")"), col.names = c("Comparison", "Variance (Group 1)",
    "Variance (Group 2)", "F-statistic", "df", "p-value", "95% CI",
    "Decision"), align = "lrrrrrrl") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 7: F-Test for Variance Comparison (= 0.05)

Comparison	Variance (Group 1)	Variance (Group 2)	F-statistic	df	p-va
PAID vs ROAMING	429.92	393.68	1.0921	22286 , 1103	4.779705e
Weekend vs Weekday	418.71	448.46	0.9337	15602 , 7787	4.433402e

Correlation and regression (uses num_data)

Correlation test between Charge Duration and Energy Provided.

Simple linear regression:

Energy Provided ~ Charge Duration.

Multiple regression adding is_weekend and a few major Location Names.

```
### Correlation and Regression Analysis Significance Level:
###   = 0.05

alpha <- 0.05
```

```r
#----------------------------------------
# 1. Correlation Test: Charge Duration vs Energy Provided
#----------------------------------------

cat("===== CORRELATION TEST =====\n")
```

```
## ===== CORRELATION TEST =====
```

```r
cat("H0:   = 0 (No linear correlation)\n")
```

```
## H0:   = 0 (No linear correlation)
```

```r
cat("Ha:    0 (Linear correlation exists)\n")
```

```
## Ha:    0 (Linear correlation exists)
```

```r
cat("  =", alpha, "\n\n")
```

```
##   = 0.05
```

```r
# Pearson correlation test
cor_test <- cor.test(ev_data$`Charge Duration (min)`, ev_data$`Energy Provided (kWh)`,
    method = "pearson", conf.level = 1 - alpha)

cat("--- Pearson Correlation Test ---\n")
```

```
## --- Pearson Correlation Test ---
```

```r
cat("Correlation coefficient (r):", round(cor_test$estimate,
    4), "\n")
```

```
## Correlation coefficient (r): 0.2319
```

```r
cat("t-statistic:", round(cor_test$statistic, 4), "\n")
```

```
## t-statistic: 36.4614
```

```r
cat("Degrees of freedom:", cor_test$parameter, "\n")
```

```
## Degrees of freedom: 23389
```

```r
cat("p-value:", format(cor_test$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 3.754562e-283
```

```r
cat("95% CI for r:", round(cor_test$conf.int[1], 4), "to", round(cor_test$conf.int[2],
    4), "\n")
```

```
## 95% CI for r: 0.2197 to 0.244
```

```r
cat("Decision:", ifelse(cor_test$p.value < alpha, "Reject H0 - Significant correlation e
    "Fail to reject H0 - No significant correlation"), "\n")
```

```
## Decision: Reject H0 - Significant correlation exists
```

```r
# Interpretation of correlation strength
r_value <- abs(cor_test$estimate)
strength <- case_when(r_value < 0.3 ~ "Weak", r_value < 0.7 ~
    "Moderate", TRUE ~ "Strong")
cat("Correlation strength:", strength, "\n")
```

```
## Correlation strength: Weak
```

```r
# Coefficient of determination
r_squared <- cor_test$estimate^2
cat("R² (coefficient of determination):", round(r_squared, 4),
    paste0("(", round(r_squared * 100, 2), "% of variance explained)\n"))
```

```
## R² (coefficient of determination): 0.0538 (5.38% of variance explained)
```

```r
# Spearman correlation (non-parametric alternative)
cor_test_spearman <- cor.test(ev_data$`Charge Duration (min)`,
    ev_data$`Energy Provided (kWh)`, method = "spearman")

cat("\n--- Spearman Correlation (Non-parametric) ---\n")
```

```
##
## --- Spearman Correlation (Non-parametric) ---
```

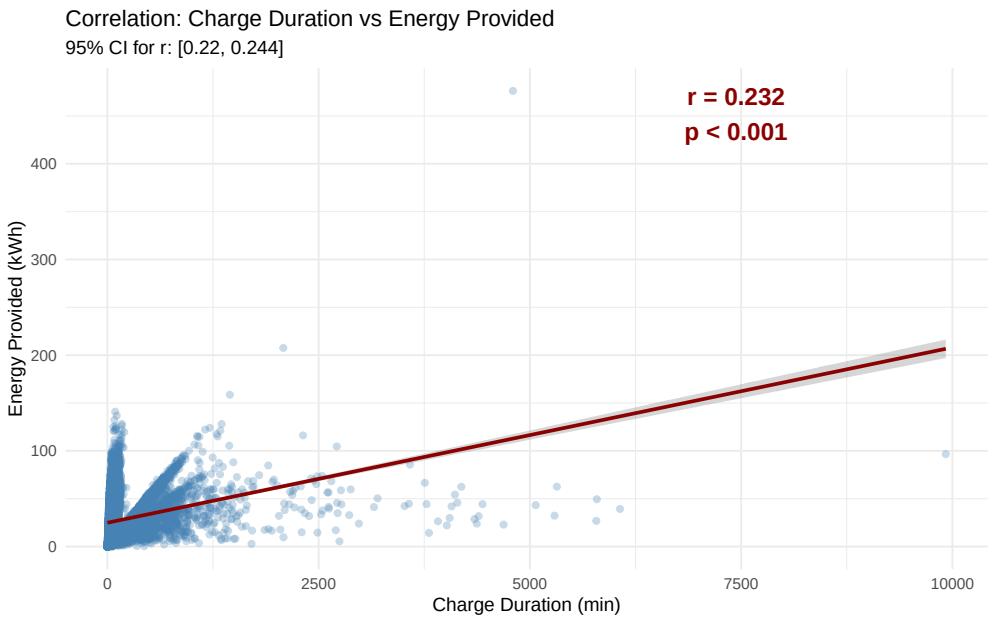
```r
cat("Correlation coefficient ( ):", round(cor_test_spearman$estimate,
    4), "\n")
```

```
## Correlation coefficient ( ): 0.4389
```

```r
cat("p-value:", format(cor_test_spearman$p.value, scientific = TRUE),
    "\n")
```

```
## p-value: 0e+00
```

```r
# Scatterplot with correlation
ggplot(ev_data, aes(x = `Charge Duration (min)`, y = `Energy Provided (kWh)`)) +
    geom_point(alpha = 0.3, color = "steelblue") + geom_smooth(method = "lm",
    se = TRUE, color = "darkred", linewidth = 1) + annotate("text",
    x = max(ev_data$`Charge Duration (min)`, na.rm = TRUE) *
        0.75, y = max(ev_data$`Energy Provided (kWh)`, na.rm = TRUE) *
        0.95, label = paste0("r = ", round(cor_test$estimate,
        3), "\np < 0.001"), size = 5, fontface = "bold", color = "darkred") +
    labs(title = "Correlation: Charge Duration vs Energy Provided",
        subtitle = paste("95% CI for r: [", round(cor_test$conf.int[1],
            3), ", ", round(cor_test$conf.int[2], 3), "]", sep = ""),
        x = "Charge Duration (min)", y = "Energy Provided (kWh)") +
    theme_minimal()
```



```r
#----------------------------------------
# 2. Simple Linear Regression
#----------------------------------------

cat("\n\n===== SIMPLE LINEAR REGRESSION =====\n")
```

```
##
##
## ===== SIMPLE LINEAR REGRESSION =====
```

```r
cat("Model: Energy Provided =  0 +  1(Charge Duration) +  \n\n")
```

```
## Model: Energy Provided =  0 +  1(Charge Duration) +
```

```r
# Fit simple linear regression
simple_model <- lm(`Energy Provided (kWh)` ~ `Charge Duration (min)`,
    data = ev_data)
simple_summary <- summary(simple_model)

cat("--- Model Summary ---\n")
```

```
## --- Model Summary ---
```

```r
print(simple_summary)
```

```
##
## Call:
## lm(formula = `Energy Provided (kWh)` ~ `Charge Duration (min)`,
##     data = ev_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -110.09  -16.79   -2.86   12.92  363.26
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             2.490e+01  1.471e-01  169.33   <2e-16 ***
## `Charge Duration (min)` 1.832e-02  5.025e-04   36.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.14 on 23389 degrees of freedom
## Multiple R-squared:  0.05378,    Adjusted R-squared:  0.05374
## F-statistic:  1329 on 1 and 23389 DF,  p-value: < 2.2e-16
```

```r
# Extract key statistics
cat("\n--- Key Statistics ---\n")
```

```
##
## --- Key Statistics ---
```

```r
cat("Intercept ( 0):", round(coef(simple_model)[1], 4), "\n")
```

```
## Intercept ( 0): 24.9007
```

```r
cat("Slope ( 1):", round(coef(simple_model)[2], 4), "\n")
```

```
## Slope ( 1): 0.0183
```

```r
cat("R²:", round(simple_summary$r.squared, 4), "\n")
```

```
## R²: 0.0538
```

```r
cat("Adjusted R²:", round(simple_summary$adj.r.squared, 4),
    "\n")
```

```
## Adjusted R²: 0.0537
```

```r
cat("Residual Std Error:", round(simple_summary$sigma, 4), "\n")
```

```
## Residual Std Error: 20.1395
```

```r
cat("F-statistic:", round(simple_summary$fstatistic[1], 4), "\n")
```

```
## F-statistic: 1329.436
```

```r
# Confidence intervals for coefficients
cat("\n--- 95% Confidence Intervals for Coefficients ---\n")
```

```
##
## --- 95% Confidence Intervals for Coefficients ---
```

```r
print(confint(simple_model, level = 1 - alpha))
```

```
##                                2.5 %      97.5 %
## (Intercept)             24.61243553 25.18891198
## `Charge Duration (min)`  0.01733531  0.01930499
```

```r
# Interpretation
cat("\n--- Interpretation ---\n")
```

```
##
## --- Interpretation ---
```

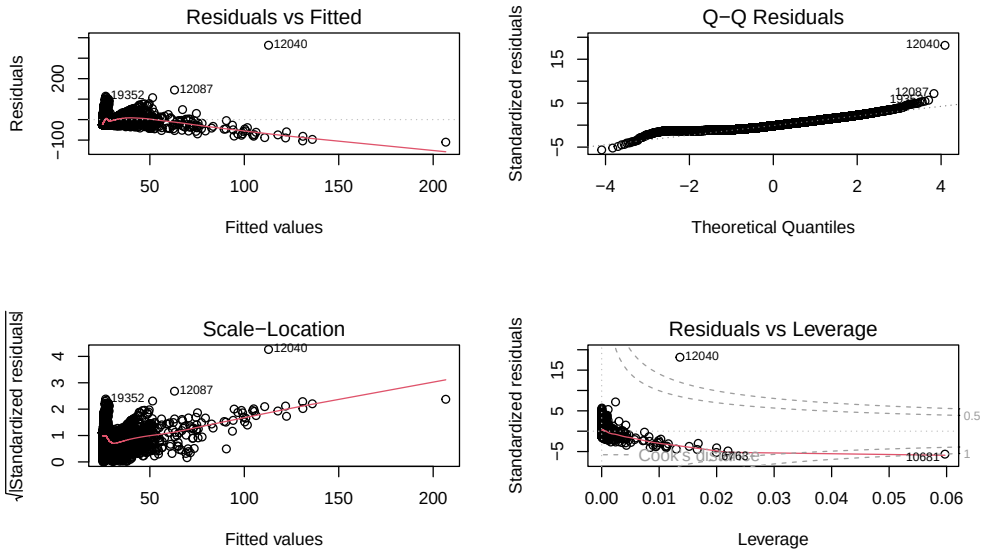
```r
cat("For every 1 minute increase in Charge Duration,\n")
```

```
## For every 1 minute increase in Charge Duration,
```

```r
cat("Energy Provided increases by", round(coef(simple_model)[2],
    4), "kWh on average.\n")
```

```
## Energy Provided increases by 0.0183 kWh on average.
```

```r
# Diagnostic plots
par(mfrow = c(2, 2))
plot(simple_model)
```



```r
par(mfrow = c(1, 1))

#----------------------------------------
# 3. Multiple Linear Regression
#----------------------------------------

cat("\n\n===== MULTIPLE LINEAR REGRESSION =====\n")
```

```
##
##
## ===== MULTIPLE LINEAR REGRESSION =====
```

```r
cat("Model: Energy Provided = 0 + 1(Charge Duration) + 2(is_weekend) + 3(Location) +
```

```
## Model: Energy Provided = 0 + 1(Charge Duration) + 2(is_weekend) + 3(Location) +
```

```r
# Prepare data: select top 4 locations for cleaner model
top_4_locations <- ev_data %>%
    count(`Location Name`, sort = TRUE) %>%
    head(4) %>%
    pull(`Location Name`)

ev_regression <- ev_data %>%
    filter(`Location Name` %in% top_4_locations) %>%
    mutate(Location = factor(`Location Name`), is_weekend = as.factor(is_weekend))

cat("Locations included in model:\n")
```

```
## Locations included in model:
```

```r
print(top_4_locations)
```

```
## [1] "Delancey and Essex Municipal Parking Garage"
## [2] "Court Square Municipal Parking Garage"
## [3] "Queens Borough Hall Municipal Parking Garage"
## [4] "Jerome 190th Street Municipal Parking Garage"
```

```r
cat("\nSample size:", nrow(ev_regression), "\n\n")
```

```
##
## Sample size: 21447
```

```r
# Fit multiple regression
multiple_model <- lm(`Energy Provided (kWh)` ~ `Charge Duration (min)` +
    is_weekend + Location, data = ev_regression)
multiple_summary <- summary(multiple_model)

cat("--- Model Summary ---\n")
```

```
## --- Model Summary ---
```

```r
print(multiple_summary)
```

```
##
## Call:
## lm(formula = `Energy Provided (kWh)` ~ `Charge Duration (min)` +
##     is_weekend + Location, data = ev_regression)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -113.27  -16.35   -2.19   12.36  361.35
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       2.929e+01  2.450e-01
## `Charge Duration (min)`                           1.852e-02  5.224e-04
## is_weekendTRUE                                   -2.303e-01  2.875e-01
## LocationDelancey and Essex Municipal Parking Garage -3.236e+00  3.023e-01
## LocationJerome 190th Street Municipal Parking Garage -1.547e+01  7.261e-01
## LocationQueens Borough Hall Municipal Parking Garage -6.729e+00  4.127e-01
##                                                   t value Pr(>|t|)
## (Intercept)                                       119.592   <2e-16 ***
## `Charge Duration (min)`                            35.457   <2e-16 ***
## is_weekendTRUE                                     -0.801    0.423
## LocationDelancey and Essex Municipal Parking Garage -10.706   <2e-16 ***
## LocationJerome 190th Street Municipal Parking Garage -21.299   <2e-16 ***
## LocationQueens Borough Hall Municipal Parking Garage -16.304   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.89 on 21441 degrees of freedom
## Multiple R-squared:  0.07819,    Adjusted R-squared:  0.07797
## F-statistic: 363.7 on 5 and 21441 DF,  p-value: < 2.2e-16
```

```r
# Extract key statistics
cat("\n--- Key Statistics ---\n")
```

```
##
## --- Key Statistics ---
```

```r
cat("R²:", round(multiple_summary$r.squared, 4), "\n")
```

```
## R²: 0.0782
```

```r
cat("Adjusted R²:", round(multiple_summary$adj.r.squared, 4),
    "\n")
```

```
## Adjusted R²: 0.078
```

```r
cat("Residual Std Error:", round(multiple_summary$sigma, 4),
    "\n")
```

```
## Residual Std Error: 19.8935
```

```r
cat("F-statistic:", round(multiple_summary$fstatistic[1], 4),
    "on", multiple_summary$fstatistic[2], "and", multiple_summary$fstatistic[3],
    "df\n")
```

```
## F-statistic: 363.7158 on 5 and 21441 df
```

```r
cat("p-value:", format(pf(multiple_summary$fstatistic[1], multiple_summary$fstatistic[2]
    multiple_summary$fstatistic[3], lower.tail = FALSE), scientific = TRUE),
    "\n")
```

```
## p-value: 0e+00
```

```r
# Confidence intervals for coefficients
cat("\n--- 95% Confidence Intervals for Coefficients ---\n")
```

```
##
## --- 95% Confidence Intervals for Coefficients ---
```

```r
print(confint(multiple_model, level = 1 - alpha))
```

```
##                                                                 2.5 %       97.5 %
## (Intercept)                                                28.8144657  29.77472735
## `Charge Duration (min)`                                     0.0175000   0.01954802
## is_weekendTRUE                                             -0.7938963   0.33326579
## LocationDelancey and Essex Municipal Parking Garage        -3.8289274  -2.64391676
## LocationJerome 190th Street Municipal Parking Garage      -16.8892381 -14.04270565
## LocationQueens Borough Hall Municipal Parking Garage       -7.5377685  -5.91990148
```

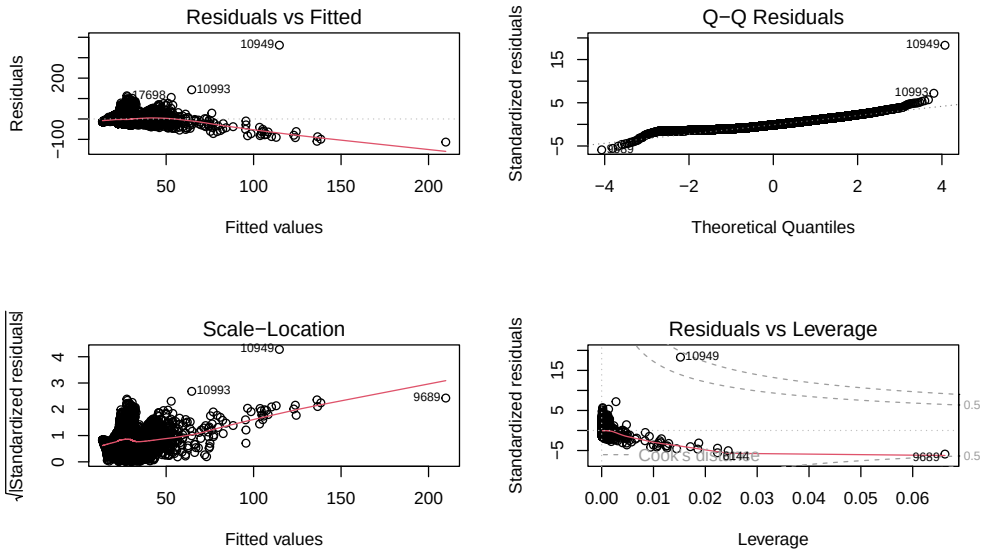
```r
# ANOVA table for multiple regression
cat("\n--- ANOVA Table ---\n")
```

```
##
## --- ANOVA Table ---
```

```r
print(anova(multiple_model))
```

```
## Analysis of Variance Table
##
## Response: Energy Provided (kWh)
##                            Df  Sum Sq Mean Sq   F value Pr(>F)
## `Charge Duration (min)`     1  470458  470458 1188.7762 <2e-16 ***
## is_weekend                  1      30      30    0.0755 0.7834
## Location                    3  249214   83071  209.9092 <2e-16 ***
## Residuals               21441 8485265     396
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Diagnostic plots
par(mfrow = c(2, 2))
plot(multiple_model)
```




```
par(mfrow = c(1, 1))

#----------------------------------------
# 4. Model Comparison
#----------------------------------------

cat("\n\n===== MODEL COMPARISON =====\n")
```

```
##
##
## ===== MODEL COMPARISON =====
```

```
# Compare R² values
cat("--- R² Comparison ---\n")
```

```
## --- R² Comparison ---
```

```
cat("Simple Model R²:", round(simple_summary$r.squared, 4),
    "\n")
```

```
## Simple Model R²: 0.0538
```

```
cat("Multiple Model R²:", round(multiple_summary$r.squared,
    4), "\n")
```

```
## Multiple Model R²: 0.0782
```

```
cat("Improvement:", round((multiple_summary$r.squared - simple_summary$r.squared) *
    100, 2), "percentage points\n")
```

```
## Improvement: 2.44 percentage points
```

```
# Compare adjusted R²
cat("\n--- Adjusted R² Comparison ---\n")
```

```
##
## --- Adjusted R² Comparison ---
```

```
cat("Simple Model Adj R²:", round(simple_summary$adj.r.squared,
    4), "\n")
```

```
## Simple Model Adj R²: 0.0537
```

```
cat("Multiple Model Adj R²:", round(multiple_summary$adj.r.squared,
    4), "\n")
```

```
## Multiple Model Adj R²: 0.078
```

```
# AIC and BIC comparison (need to fit on same data)
simple_model_subset <- lm(`Energy Provided (kWh)` ~ `Charge Duration (min)`,
    data = ev_regression)

cat("\n--- AIC/BIC Comparison ---\n")
```

```
##
## --- AIC/BIC Comparison ---
```

```
cat("Simple Model - AIC:", round(AIC(simple_model_subset), 2),
    ", BIC:", round(BIC(simple_model_subset), 2), "\n")
```

```
## Simple Model - AIC: 189754.7 , BIC: 189778.6
```

```r
cat("Multiple Model - AIC:", round(AIC(multiple_model), 2), ", BIC:",
    round(BIC(multiple_model), 2), "\n")
```

```
## Multiple Model - AIC: 189141.8 , BIC: 189197.6
```

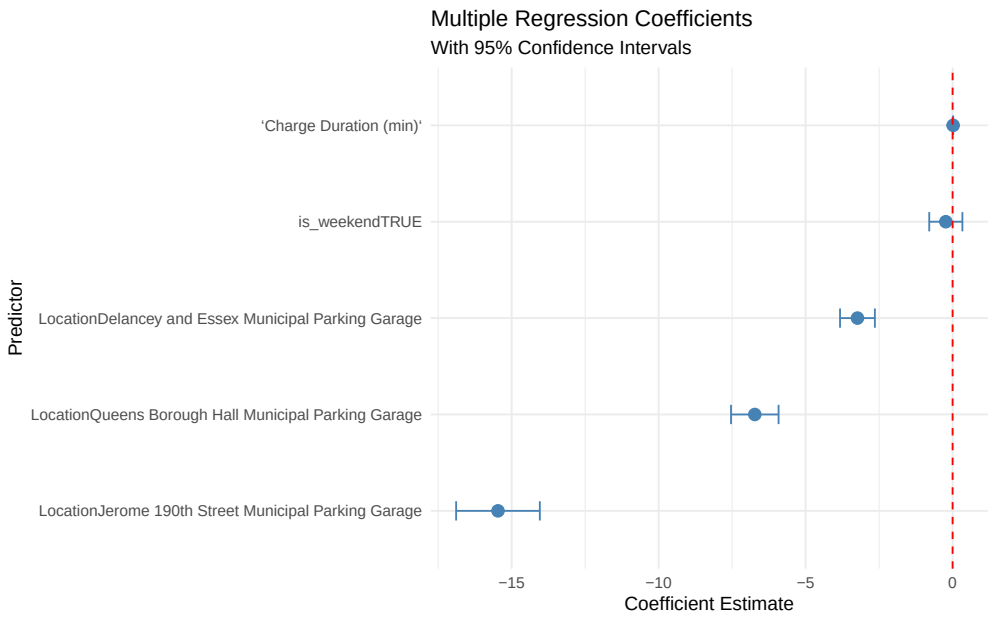
```r
cat("Lower values indicate better fit\n")
```

```
## Lower values indicate better fit
```

```r
#----------------------------------------
# 5. Coefficient Visualization
#----------------------------------------

# Extract coefficients for visualization
coef_df <- data.frame(Term = names(coef(multiple_model)), Estimate = coef(multiple_model
    CI_lower = confint(multiple_model)[, 1], CI_upper = confint(multiple_model)[,
        2]) %>%
    filter(Term != "(Intercept)")  # Remove intercept for cleaner plot

# Coefficient plot
ggplot(coef_df, aes(x = reorder(Term, Estimate), y = Estimate)) +
    geom_point(size = 3, color = "steelblue") + geom_errorbar(aes(ymin = CI_lower,
    ymax = CI_upper), width = 0.2, color = "steelblue") + geom_hline(yintercept = 0,
    linetype = "dashed", color = "red") + coord_flip() + labs(title = "Multiple Regressi
    subtitle = "With 95% Confidence Intervals", x = "Predictor",
    y = "Coefficient Estimate") + theme_minimal()
```



```r
#----------------------------------------
# 6. Summary Tables
#----------------------------------------

# Correlation summary
correlation_summary <- data.frame(Method = c("Pearson", "Spearman"),
    Coefficient = c(round(cor_test$estimate, 4), round(cor_test_spearman$estimate,
        4)), p_value = c(format(cor_test$p.value, scientific = TRUE),
        format(cor_test_spearman$p.value, scientific = TRUE)),
    CI_95 = c(paste0("(", round(cor_test$conf.int[1], 4), ", ",
        round(cor_test$conf.int[2], 4), ")"), "N/A"), Decision = c(ifelse(cor_test$p.val
        alpha, "Significant", "Not Significant"), ifelse(cor_test_spearman$p.value <
        alpha, "Significant", "Not Significant")))

kable(correlation_summary, caption = paste("Correlation Test Summary (  =",
    alpha, ")"), col.names = c("Method", "Coefficient", "p-value",
    "95% CI", "Decision"), align = "lrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```

Table 8: Correlation Test Summary (= 0.05)

	Method	Coefficient	p-value	95% CI	Decision
cor	Pearson	0.2319	3.754562e-283	(0.2197, 0.244)	Significant
rho	Spearman	0.4389	0e+00	N/A	Significant

```r
# Regression model comparison
model_comparison <- data.frame(Model = c("Simple (Duration only)",
    "Multiple (Duration + Weekend + Location)"), R_squared = c(round(simple_summary$r.sq
    4), round(multiple_summary$r.squared, 4)), Adj_R_squared = c(round(simple_summary$ad
    4), round(multiple_summary$adj.r.squared, 4)), RSE = c(round(simple_summary$sigma,
    4), round(multiple_summary$sigma, 4)), AIC = c(round(AIC(simple_model_subset),
    2), round(AIC(multiple_model), 2)), BIC = c(round(BIC(simple_model_subset),
    2), round(BIC(multiple_model), 2)))

kable(model_comparison, caption = "Regression Model Comparison",
    col.names = c("Model", "R²", "Adj R²", "Residual SE", "AIC",
        "BIC"), align = "lrrrrr") %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
        full_width = FALSE, position = "center")
```